Sorting Over Wildfire Risk Information*

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Abstract

We study how information about climate risk changes household beliefs, housing demand, and residential sorting in equilibrium. We use the introduction of a wildfire risk disclosure policy in California as a shock to beliefs about wildfire risk that leaves the underlying physical hazard unchanged, allowing us to identify how changes in beliefs affect location choices and housing prices. We embed a Bayesian model of belief updating in a structural model of neighborhood choice in which households have heterogeneous preferences over housing prices and wildfire risk. Our estimates indicate that before the policy, households held diffuse beliefs about neighborhood wildfire risk, and disclosure widened the gap in believed annual burn probability between disclosed and non-disclosed neighborhoods by 0.47 percentage points. Counterfactual simulations that remove the policy show that this change in beliefs re-sorts households by their sensitivity to wildfire risk: mortgage-financed buyers of primary homes, the most risk-sensitive group, shift away from disclosed neighborhoods, while buyers of non-primary homes, especially cash buyers, move in, and prices in disclosed neighborhoods fall relative to non-disclosed neighborhoods. Relative to the no-disclosure counterfactual, the gap in average annual fire exposure between the neighborhoods chosen by cash buyers of non-primary homes and mortgage-financed buyers of primary homes, from the year of purchase through 2023, increases by 29%. Alternative counterfactual exercises show that a finer-grained risk score could achieve a similar or larger reallocation of risk with substantially less household re-sorting. Our findings show that information can reshape beliefs about wildfire risk, housing market outcomes, and the incidence of wildfire exposure, even when the underlying physical hazard is unchanged.

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1 Introduction

The allocation of climate risk across households depends not only on the spatial distribution of physical hazards, but also on households' beliefs about those hazards when making location decisions. When households have imperfect beliefs about risk, providing new information can change where they choose to live, inducing re-sorting across space and changes in equilibrium housing prices. These equilibrium responses can, in turn, change which households ultimately bear climate risk. Understanding how households trade off wildfire risk against other amenities, and how providing risk information alters households' decisions about where to live, is important to understanding this re-allocation. In this paper, we study how information about wildfire risk updates household beliefs and how these belief changes reshape residential sorting and the equilibrium allocation of risk across households.

In 2020, California enacted legislation that requires sellers to disclose wildfire risk information to homebuyers during the home sales process. The policy introduced standardized information about wildfire exposure during the home buying process without changing the underlying physical risk. In this paper, we use this policy reform to examine how climate risk information affects housing market outcomes and residential sorting. We find that households re-sort in response to the risk information policy. As households update their beliefs about wildfire risk, more risk-sensitive buyers sort away from disclosed areas, while less risk-sensitive buyers sort in.

A growing literature shows that households often underestimate exposure to climate risks such as flooding and wildfire, but update their beliefs in response to new information (Bakkensen and Barrage, 2022; Gibson and Mullins, 2020; Weill, 2026). Consistent with this idea, prior work finds that wildfire risk disclosure in California affects housing market outcomes, reducing prices in high-risk areas and shifting the composition of buyers away from mortgage-financed, owner-occupant purchases toward cash buyers and second-home investors, particularly in areas with more severe insurance constraints (Bass and Fogarty, 2025; Ma et al., 2024). Together, these findings suggest that disclosure induces substantial re-sorting across wildfire risk. However, reduced-form evidence cannot separately recover the beliefs and preferences that generate these sorting responses. In particular, it cannot quantify the extent to which households misperceive wildfire risk or how correcting those beliefs interacts with heterogeneous preferences for risk and housing prices to generate re-sorting.

More broadly, reduced-form estimates of disclosure effects are inherently local and do not capture how responses propagate through the broader housing market. By design, these estimates compare areas near disclosure boundaries where underlying wildfire risk is similar, limiting their ability to predict how information would affect sorting patterns across the full distribution of risk or under alternative policy regimes. In addition, they do not recover the underlying preferences governing household location choices or account for how equilibrium price adjustments across neighborhoods amplify or dampen sorting responses. Understanding how information reshapes residential sorting therefore requires a framework that jointly models household preferences, beliefs about risk, and equilibrium housing prices.

This paper develops and estimates a structural model of neighborhood sorting in which households have heterogeneous preferences over housing prices and wildfire risk and beliefs about wildfire risk that update in response to new information. We use spatial variation in wildfire risk disclosure to identify households' prior beliefs about wildfire risk and their heterogeneous preferences for risk and housing prices. The model allows us to quantify the extent to which households update their beliefs about wildfire risk in response to disclosure and to evaluate how belief updating and equilibrium price adjustments jointly determine sorting patterns in response to climate risk information.

The model allows household preferences for price and risk to differ across buyer types defined by financing status (mortgage vs. cash financing) and intended home use (primary residence vs. non-primary use). In the absence of disclosure, buyers have beliefs about wildfire risk that are diffuse and potentially inaccurate. The disclosure policy induces households to update their beliefs: neighborhoods subject to risk disclosure are perceived to be riskier, while neighborhoods not subject to disclosure are perceived to be safer. These changes in perceived risk shift housing demand across neighborhoods and generate equilibrium price adjustments.

We estimate the model using microdata on the universe of single family home transactions in California from 2017-2023 alongside a panel of neighborhood characteristics that includes prices, wildfire risk, observed market shares, and measures of insurance market conditions. We first construct neighborhood housing price indices using a hedonic model that combines rich property characteristics with machine learning methods. We then estimate heterogeneous household preferences for housing prices and wildfire risk using maximum likelihood, leveraging observed neighborhood choices to recover how preferences and beliefs differ across buyers. To address the endogeneity of housing prices to local amenities, we instrument

for neighborhood prices using a shift-share approach that combines cross-neighborhood differences in buyer type preferences with national trends in the composition buyer types.

Estimating the model reveals substantial heterogeneity in how buyers trade off housing prices and wildfire risk. Risk sensitivity is greatest among mortgage-financed buyers purchasing primary residences and lowest among cash and non-primary purchasers. We estimate that cash-financed buyers of primary homes are about 20% less price sensitive than the other buyer types. We also find that households have substantial uncertainty surrounding their prior beliefs, and this uncertainty leads households to update substantially in response to the disclosure policy. Our estimates suggest that disclosure drives a divergence in beliefs about wildfire risk between disclosed and non-disclosed neighborhoods of 0.47 percentage points of annual burn probability.

To quantify the effects of the disclosure policy on sorting and equilibrium prices, we conduct a counterfactual exercise in which we shut off the disclosure policy in the post-policy period (2021–2023), holding estimated preferences and unobserved amenities fixed. Disclosure substantially reallocates buyer types across neighborhoods. Holding housing supply fixed, the policy raises the share of cash buyers of non-primary homes, the type least sensitive to wildfire risk, in disclosed neighborhoods by 4.2 percentage points and the share of mortgage-financed buyers of non-primary homes by 3.0 percentage points, while lowering the share of mortgage-financed buyers of primary homes by 7.4 percentage points. Effects in neighborhoods not subject to disclosure are smaller and run in the opposite direction. Prices in disclosed neighborhoods fall by 0.65% and prices in non-disclosed neighborhoods rise by 0.14%. These results change little in a longer-run counterfactual that allows housing supply to respond, using the neighborhood supply elasticities from Baum-Snow and Han (2024): the sorting effects are nearly identical, and prices in disclosed neighborhoods fall by 0.56%.

We then use the model to evaluate alternative disclosure designs. We vary how risk is conveyed, comparing California's binary designation with a binary label drawn from the hazard map and with a decile score of burn probability, and whether buyers observe the label without noise. Design matters both for how much risk is reallocated and how much disruption that reallocation requires. Making the binary label more precise reallocates less risk while changing more households' choices, because the coarse label changes beliefs in many neighborhoods whose risk it misstates. A precise decile score, by contrast, raises the gap in exposure between the least and most risk-sensitive buyers by 42%, compared with 25% under

the current policy, while changing the choices of 0.9% of households rather than 2.3%. Finally, we find that making housing supply more elastic in low-risk neighborhoods does little to change the effects of disclosure, because the changes in demand induced by the policy are not sufficiently large to create a large quantity response in newly-desirable low risk neighborhoods.

A growing literature studies how climate risk information enters housing markets through disclosure policies, insurance pricing, and platform-based information provision. Empirical work shows that formal disclosures and information shocks—such as wildfire hazard disclosures (Ma et al., 2024), regulatory flood maps (Hino and Burke, 2021), and large-scale platform experiments (Fairweather et al., 2023)—affect housing search behavior, transaction outcomes, and equilibrium prices, although capitalization is often incomplete or context-dependent, consistent with shifting salience (Bin and Landry, 2013). Recent work expands beyond price effects to examine additional margins of adjustment, including financing and insurance: Bass and Fogarty (2025) show that wildfire risk disclosure shifts transactions toward cash purchases and reduces owner-occupancy, while Collier et al. (2023) documents substantial declines in insurance take-up following premium increases. While this literature establishes that climate risk information affects housing market behavior, less is known about the household beliefs and preferences that generate these responses or how they aggregate in equilibrium. We recover these underlying beliefs and preferences and use them to characterize how information provision reshapes housing demand and residential sorting across the broader market.

A second literature focuses on how individuals perceive climate risk and how those perceptions shape economic behavior, emphasizing belief formation, updating, and heterogeneity. Early work shows that direct experience with extreme events can update perceived risk, as reflected in housing market outcomes (Kousky, 2010), while more recent studies document that risk perceptions respond to both informational signals (e.g., flood maps) and lived experience (Gibson and Mullins, 2020). A central finding is that beliefs are often systematically biased or misinformed: (Mulder, 2024) shows that incorrect risk classification can distort insurance and adaptation decisions with welfare consequences, and (Bakkensen and Barrage, 2022) documents substantial heterogeneity and underestimation of flood risk that helps explain mixed evidence on price capitalization. Complementary experimental evidence finds that information provision can shift stated beliefs about climate risk but does not always translate into behavioral change (Deryugina and Shurchkov, 2016). We build on this work by recovering households' perceptions of wildfire risk directly from revealed

residential choices. The shock to beliefs induced by the disclosure policy allows us to disentangle beliefs from heterogeneity in households' preferences for wildfire risk as separate channels that impact household residential choices.

A third strand of literature examines how climate risk shapes residential sorting and migration decisions, building on a foundational literature estimating equilibrium models of location choice (Epple and Sieg, 1999; Epple et al., 2001; Bayer and Timmins, 2007) and heterogeneous preferences for neighborhood attributes (Bayer et al., 2007; Ioannides and Zabel, 2008). Applied work incorporates climate risk into these frameworks, showing that households sort across locations based on exposure to environmental risk and that these responses are highly heterogeneous: Bakkensen and Ma (2020) find differential sorting over flood risk by income and race with important distributional implications. We extend this literature by allowing the risk households perceive when choosing where to live to differ from underlying physical risk and to change with new information. Incorporating belief updating into an equilibrium model of residential sorting allows information itself to reshape the spatial allocation of households, even when the underlying distribution of physical risk is unchanged. Our model builds on prior work by Ma (2019) and Barahona et al. (2023) that incorporates endogenous learning about attributes into discrete choice demand models.

We proceed as follows. We begin with institutional background describing the rollout of the disclosure requirement and the information contained in the natural hazard disclosure form. We next introduce the model of belief updating in response to information provision, and embed households' beliefs about wildfire risk in to a model of neighborhood choice. We then turn to describe the data we use to estimate the model. Next, we describe the estimation procedure and identification of key parameters and present the accompanying results. We then conduct a number of counterfactual policy simulations to evaluate the effects of the disclosure policy and consider alternative policy designs. We close with a discussion of implications, limitations, and conclusions.

2 Institutional Background

Wildfire risk has become an increasingly important consideration for households in California, where annual wildfire burned area increased roughly fivefold between 1972 and 2018 (Williams et al., 2019). Households choosing where to live must weigh the benefits of fire-prone locations, such as the amenity

value of natural beauty, against the risks of property damage, financial loss, and disruption. These risks vary substantially across space, making wildfire exposure a key dimension along which neighborhoods differ.

Households can respond to wildfire risk along several margins, including investments in property-level mitigation, insurance coverage, and residential location. These responses are shaped by both private and public institutions. Building standards can substantially reduce wildfire losses, including through spillovers that reduce structure-to-structure fire spread (Baylis and Boomhower, 2026), while public wildfire suppression can subsidize development in high-risk areas by shifting some of the costs of wildfire protection to the government (Baylis and Boomhower, 2023). The financial consequences of wildfire also extend beyond direct property losses: wildfire damage can disrupt mortgage repayment and expose gaps between insurance settlements and rebuilding costs (Biswas et al., 2023). Insurance markets provide another important mechanism for allocating wildfire risk. Insurers increasingly use granular measures of wildfire exposure to classify and price properties (Boomhower et al., 2024; Boomhower and Fowlie, 2026), while mortgage lenders can respond to residual disaster risk by changing the terms on which households can finance homes in risky areas (Sastry, 2026). These institutional responses affect both the costs households face from living in high-risk locations and which households are able or willing to bear those costs.

Despite the importance of wildfire risk, households may not fully perceive the risks they face when making these decisions. Limited information about both wildfire exposure and the returns to self-protection can inhibit private adaptation (Boomhower et al., 2023; Baylis and Boomhower, 2026). Moreover, insurance prices may not effectively communicate underlying risk to households. Mulder and Kousky (2023) show that risk-based insurance pricing may have limited effects on adaptation when the risk information underlying those prices is not directly communicated to households. Information frictions are particularly important for residential location decisions because purchasing a home is a long-lived investment that determines a household's exposure to wildfire risk, insurance costs, and potential property losses.

California's wildfire risk disclosure policy introduced standardized information about wildfire exposure into housing transactions. Beginning in 2021, California added new fields to its preexisting natural hazard disclosure form indicating whether a property is located in a designated wildfire hazard zone based on Fire Hazard Severity Zone (FHSZ) maps created by the California Department of Forestry and Fire Prevention.[1] Properties located in areas designated by Cal Fire as "Very High" fire hazard severity zones, as well as

[1]Figure A1 displays the updated Natural Hazard Disclosure form that includes the new wildfire hazard fields

areas designated as "High" or "Moderate" that lie within state-managed fire protection zones, are required to check "Yes" on the relevant wildfire fields in the Natural Hazard Disclosure Form.

The information provided in the disclosure is coarse, indicating exposure to mapped risk zones rather than property-specific risk, and includes a general notice that buyers may face challenges obtaining homeowners insurance. The disclosure does not change the underlying risk, but provides buyers with new information about that risk at the time of purchase. This distinction between underlying risk and information about risk is central to our empirical setting: the policy changes the information available to households without directly changing the physical hazard they face.

3 Model

To study how wildfire risk information disclosure shapes residential sorting, we develop a discrete choice model of neighborhood demand in which households have beliefs about wildfire risk, and the risk disclosure policy induces a Bayesian updating of households' beliefs about wildfire risk.

We adapt the belief updating model from Barahona et al. (2023), who model the impact of a food labeling policy on consumer demand for breakfast cereals, to better fit our setting. In our context, wildfire risk disclosure is a noisy, but still informative, signal of underlying wildfire risk. This causes households to shade down the extent to which they update their beliefs compared to the fully informative case.

In our setup, wildfire risk disclosure ($D_j \in \{0, 1\}$) is a noisy but informative signal of underlying wildfire risk. Upon signal receipt, households update their beliefs in a Bayesian manner, with the extent of the updating moderated by the amount of noise in the disclosure signal.

The motivation for this approach comes from Bass and Fogarty (2025), which documents that the disclosure policy not only reduced home prices in disclosed areas, but also shifted the composition of buyers toward cash purchases and away from primary home use. These changes in buyer composition are difficult to explain through a full-information model, and are instead consistent with an information story in which disclosure makes wildfire risk more salient at the point of purchase, causing different types of buyers to update their behavior differently. This paper directly models that information channel by treating disclosure as a trigger that induces households to update their beliefs about wildfire risk.

We close the model with a housing supply specification that governs how changes in demand are

allocated between price adjustments and quantity responses. In equilibrium, neighborhood prices adjust until supply equals demand in each neighborhood. The remainder of this section describes the demand and supply sides of the model in turn.

3.1 Neighborhood Demand

We propose a neighborhood choice model that treats neighborhoods as differentiated products, as in Bayer et al. (2007). Households buying a house in period t solve the following discrete choice problem:

$$\max_{j \in J_t} V_{ijt} \tag{1}$$

The choice set J_t consists of all neighborhoods defined by the intersection of ZCTAs with the risk disclosure area, plus an outside option of relocating outside of California. Equation 2 gives the indirect utility of household i choosing neighborhood j in period t.

$$V_{ijt} = \theta^P_{z(i)} P_{jt} + \theta^R_{z(i)} E_t[r_{jt}|D_j] + X_{jt}\beta + \xi_{jt} + \varepsilon_{ijt} \tag{2}$$

P_{jt} is a price index for the price of housing services in neighborhood j in year t, $E_t[r_{jt}|D_j]$ is the mean of household i's belief about wildfire risk in neighborhood j in period t, and X_{jt} are observable neighborhood characteristics. ξ_{jt} is an unobserved (to the econometrician) neighborhood amenity. ε_{ijt} is an iid taste shock distributed according to the Type-1 extreme value distribution. We normalize the scale of the taste shocks by setting $\sigma_\varepsilon = 1$.

Price sensitivity varies across different buyer types Z, but is constant over time in the following way:

$$\theta^P_{z(i)} = \bar{\theta}^P + \sum_{k=1}^{3} z_{ik}\theta^P_k$$

Here, z_{ik} is a set of indicators for whether household i belongs to types $k \in \{1, 2, 3\}$. Likewise, risk aversion also varies by buyer types Z, but is consistent with respect to wildfire risk beliefs $E_t[r_j|D_j]$,

$$\theta^R_{z(i)} = \bar{\theta}^R + \sum_{k=1}^{3} z_{ik}\theta^R_k$$

Observable household heterogeneity follows buyer types Z, which are given by the intersection of two characteristics: mortgage financing status (cash versus mortgage financed purchase) and home purchase type (owner-occupied versus non-owner-occupied). This intersection creates four types of buyers, which are detailed in Table 1. We treat mortgage-financed buyers of primary homes as the reference category for the heterogeneous preferences over price and risk because they are the modal buyer type, accounting for an outright majority of home purchases in the state of California across our sample period.

3.1.1 Prior Beliefs and Learning from Fires

Let r_j denote the annual burn probability of neighborhood j: the probability that a given home in j burns in a wildfire in a given year. Households do not observe r_j, but instead learn about it from three sources. The first is the baseline level of wildfire risk in the neighborhood, which we measure as R_j, the USFS estimated burn probability (Scott et al., 2024) aggregated to the neighborhood level using housing-unit weights. R_j anchors households' prior beliefs. The second is the history of wildfires in the neighborhood, and the third, beginning in 2021, is the disclosure designation D_j. This section describes households' prior beliefs and how those beliefs evolve as households observe fires. Section 3.1.3 describes how households update in response to the risk disclosure policy.

We model beliefs with a Gamma–Poisson learning model. Let f_{js} denote wildfire exposure in neighborhood j in year s, measured as the share of homes in j reached by wildfire. Fire exposure arrives according to a Poisson process with annual rate r_j.[2] We assume that r_j is constant over time, which we view as a reasonable approximation over the relatively short timespan of our estimation sample.

Parameterization. Households' prior over r_j is a Gamma distribution centered on the USFS burn probability R_j, with a single precision parameter κ common to all neighborhoods:

$$r_j \sim \Gamma(\kappa R_j,\ \kappa), \tag{3}$$

in shape–rate form. The prior has mean R_j and variance R_j/κ. Before households observe any fires, beliefs are centered on the baseline risk R_j. The parameter κ measures the number of years of fire history the prior

[2]For small r_j, the Poisson rate and the annual probability of at least one fire, $1 - e^{-r_j}$, are approximately equal.

is worth: the prior is equivalent to having observed κ years of fires with total exposure κR_j. A larger κ means households place more weight on the scientific risk estimate relative to the fires they subsequently observe.

Learning from fires. Let $F_{jt} = \sum_{s=2012}^{t-1} f_{js}$ denote cumulative fire exposure observed by a household buying in year t, and $T_t = t - 2012$ the number of years of fire history. Under constant risk assumption, F_{jt} and T_t are sufficient statistics for beliefs, and the exact timing of fires within the observed history does not matter. Because the Gamma prior is conjugate to the Poisson likelihood, the posterior is also Gamma distributed:

$$r_j \mid F_{jt}, T_t \sim \Gamma(\kappa R_j + F_{jt},\ \kappa + T_t), \tag{4}$$

with mean

$$E_t[r_j | F_{jt}, T_t] = \frac{\kappa R_j + F_{jt}}{\kappa + T_t}. \tag{5}$$

The posterior mean is a weighted average of the hazard-map risk R_j and the neighborhood's observed burn rate F_{jt}/T_t. The weight on observed fires rises as the history lengthens.

Beliefs respond to both fires and their absence, but in a highly asymmetric manner. Beliefs update sharply upwards after observing a fire, but drift down slowly in its absence. Because households' prior beliefs about risk are low (on the order of 0-3% annual burn probability), the increase in beliefs in response to a fire is an order of magnitude larger than the decline caused by the absence of a fire in a given year.

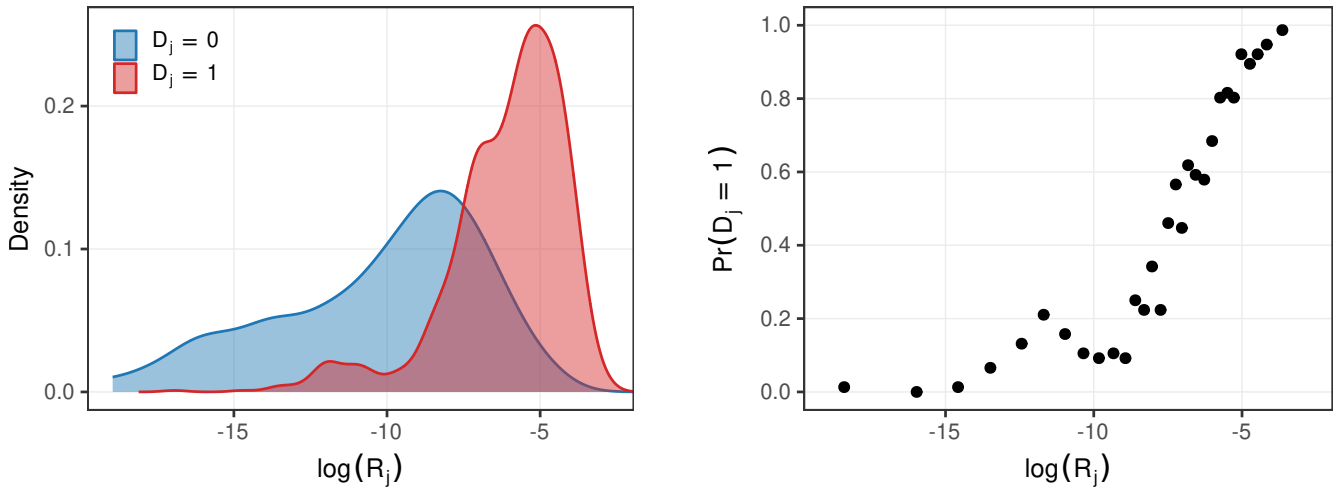
In the absence of disclosure, Equation (4) fully describes household beliefs. Beginning in 2021, it serves as the prior for the disclosure update described in Section 3.1.3.

3.1.2 Disclosure Policy

One key difference between our setting and the Barahona et al. (2023) cereal labeling paper the binary wildfire risk disclosure signal is not perfectly informative of the underlying risk level. In this setting D_j is a noisy, but informative, signal of underlying risk. The Cal Fire hazard zones do not perfectly segment neighborhoods by observable risk score; Figure 1a shows the overlap between the distribution of risk scores for disclosed (red) and non-disclosed (blue) neighborhoods.

Figure 1b shows a binned scatter plot of the conditional probability that a neighborhood receives a

disclosure designation as a function of the risk score R_j. Disclosure probability is relatively flat across low levels of neighborhood risk before increasing sharply at higher levels.



(a) Burn probability by disclosure status

(b) Probability of disclosure as a function of risk

Figure 1: Wildfire Risk and Disclosure Status

Notes. The unit of observation is the neighborhood, the intersection of a ZCTA with the FHSZ-regulated area; $D_j = 1$ denotes neighborhoods subject to the disclosure requirement from 2021. The horizontal axis in both panels is the log of the continuous risk index R_j, the mean of the USFS 2023 wildfire hazard potential raster over the neighborhood polygon divided by 1,000. Panel (a) plots kernel density estimates of $\log R_j$ separately for disclosed and non-disclosed neighborhoods. Panel (b) groups neighborhoods into 30 equal-count bins of $\log R_j$ and plots, for each bin, the share of neighborhoods with $D_j = 1$ against the bin's mean $\log R_j$. The 83 neighborhoods with a hazard score of zero, for which $\log R_j$ is undefined, are excluded, leaving 2,263 neighborhoods (989 disclosed, 1,274 not disclosed).

We model the noisy disclosure rule using a Probit model:

$$\Pr(D_j = 1) = \Phi\left(\frac{R_j - \overline{r}}{\tau}\right) \tag{6}$$

Here, $\{\overline{r}, \tau\}$ the government's disclosure "policy function". We externally estimate the policy parameters using maximum likelihood to recover cutoff and noise parameters, and take these parameters as given in the model of households' belief updating process.

The noise in the policy rule introduces additional uncertainty into the updating process and makes the disclosure signal less informative. As a result, households will shade their updating relative to the perfectly informative signal case.

3.1.3 Updating on Disclosure

Beginning in 2021, households also observe the disclosure designation D_j, which they treat as a noisy signal of r_j generated by the disclosure rule in Equation (6). Households combine the disclosure with the fire history using Bayes' rule. Under the risk stationarity assumption, the order in which buyers observe the two signals does not matter. The posterior is the fire-updated Gamma belief from Equation (4) multiplied by the likelihood of the observed designation:

$$h(r_j \mid F_{jt}, D_j) \propto g(r_j;\, \kappa R_j + F_{jt},\, \kappa + T_t)\ \Pr(D_j \mid r_j), \tag{7}$$

where g is the Gamma density associated with the fire-updated prior.

Unlike the fire update, the disclosure likelihood is not conjugate to the Gamma distribution, so the posterior is no longer Gamma. However, its mean still takes a simple form: the fire-updated prior mean times a multiplicative updating factor:

$$E[r_j \mid F_{jt}, D_j] = \underbrace{\mu_{jt}}_{\text{Prior mean}} \times \underbrace{\left(1 + \frac{\mathrm{Cov}_t\big(r_j,\, \Pr(D_j \mid r_j)\big)}{\mu_{jt}\ \Pr_t(D_j)}\right)}_{\text{Updating factor}}, \tag{8}$$

where μ_{jt} is the prior mean from equation (5) and $\Pr_t(D_j) = E_t[\Pr(D_j \mid r_j)]$ is the probability households assign to the observed designation before seeing it, given the policy function $\{\bar{r}, \tau\}$. The size of the update depends on two terms. The covariance in the numerator measures how informative the designation is about risk. Because the probability of disclosure rises with r_j, the covariance is positive for disclosed neighborhoods and negative for non-disclosed ones. Therefore, the updating factor is greater than one after a disclosure and less than one after non-disclosure. The denominator scales the update by how "surprising" the designation is: a designation that households considered unlikely moves beliefs more. Appendix C derives a closed-form expression for the updating factor under the Gamma prior and describes how it varies across neighborhoods.

The size of the update depends on the noise in the disclosure rule. A larger τ flattens $\Pr(D_j = 1 \mid r_j)$, which weakens the covariance with r_j and shrinks the update in both directions. As $\tau \to \infty$, the designation is uninformative and the covariance shrinks to zero, so beliefs remain at μ_{jt}. As $\tau \to 0$, the rule becomes a

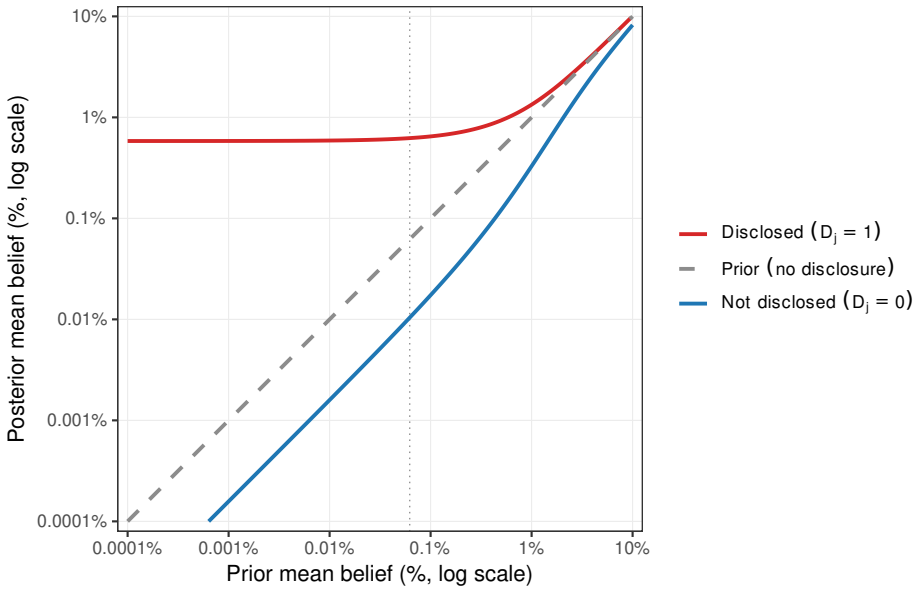


Figure 2: Posterior Beliefs After Disclosure

Notes. Posterior mean beliefs about the annual burn probability r_j in 2021 ($T_t = 9$), plotted against the fire-updated prior mean $\mu_{jt} = (\kappa R_j + F_{jt})/(\kappa + T_t)$, with κ, $\bar{r}$, and τ set to their estimated values. Both axes are on log scales. The dashed 45-degree line is the prior, which is also the belief absent disclosure. The solid curves are the posterior means after a disclosure ($D_j = 1$) and after non-disclosure ($D_j = 0$), computed from equation (8) using the closed form in Appendix C. Because the posterior depends on R_j and F_{jt} only through μ_{jt}, the curves apply to every neighborhood in 2021, with or without a fire history. The dotted vertical line marks a prior mean equal to the disclosure threshold, $e^{\bar{r}}$. The $D_j = 0$ curve falls below the plotted range for prior means under about 0.0006%.

deterministic threshold at $\bar{r}$, and the posterior a Gamma distribution truncated at $\bar{r}$. Because prior beliefs are diffuse, even a noisy rule carries substantial information about r_j.

Figure 2 illustrates the updating process. It plots the posterior means against the prior in 2021 for the two disclosure outcomes. The dashed 45 degree line is the prior. Disclosure pushes beliefs above the line and non disclosure pushes them below it. Signals that contradict the prior update beliefs the most. In neighborhoods with low prior risk, disclosure raises beliefs to nearly the same level regardless of the prior, while non-disclosure lowers beliefs by a roughly constant proportion. When prior beliefs are high, disclosure confirms the prior and shifts beliefs little.

3.1.4 Choice Probabilities and Neighborhood Demand

Households differ in their preferences only through their type k, so choice probabilities are common to all households of a given type. Writing utility as mean utility δ_{jt} plus type-specific deviations, the

probability that a type-k household chooses neighborhood j in year t takes the multinomial logit form

$$\pi_{kjt} = \frac{\exp\left(\delta_{jt} + \theta_k^P \log P_{jt} + \theta_k^R E[r_j \mid F_{jt}, D_j]\right)}{1 + \sum_{\ell \in J_t} \exp\left(\delta_{\ell t} + \theta_k^P \log P_{\ell t} + \theta_k^R E[r_\ell \mid F_{\ell t}, D_\ell]\right)}, \tag{9}$$

where J_t is the set of neighborhoods in year t and $E[r_j \mid F_{jt}, D_j]$ is the posterior mean burn probability from Section 3.1.3, measured in percentage points.[3] The model-implied market share of neighborhood j aggregates these probabilities across household types:

$$s_{jt}^D = \sum_k w_{kt}\, \pi_{kjt}, \tag{10}$$

where w_{kt} is the share of type-k households in the year-t market.

The outside option is to leave California, with utility normalized to zero; it corresponds to the 1 in the denominator of equation (9). The market size N_t is the number of households in the market for an owner-occupied home in California in year t: the number of home purchases in California plus the net number of owner households that move out of the state.[4] Observed market shares are $s_{jt} = n_{jt}/N_t$, where n_{jt} is the number of purchases in neighborhood j, and the outside share $s_{0t} = 1 - \sum_{j \in J_t} s_{jt}$ is the share of the market that leaves California. The outside option allows total demand for California housing to respond to the disclosure policy: a household that would otherwise buy in a disclosed neighborhood can leave the state instead of substituting to another California neighborhood. Without it, inside shares would sum to one, and any decline in demand for disclosed neighborhoods would have to materialize elsewhere in California.

A key step in the estimation procedure is matching the model-implied market shares s_{jt}^D to the observed shares s_{jt}. For given values of the type-specific deviations and the belief parameter κ, there exists a unique vector of mean utilities δ_t that equates the two (Berry, 1994), and it can be computed by the contraction mapping of Berry et al. (1995). This inversion delivers δ_{jt} as a function of the remaining parameters, which we describe further in Section 5.

[3] Before 2021, households do not observe D_j, and beliefs are the fire-updated mean $E[r_j \mid F_{jt}] = \frac{\kappa R_j + F_{jt}}{\kappa + T_t}$.

[4] We use the net outflow because households that move into California and buy a home are already counted among purchases. Section 4 describes the construction of N_t.

3.2 Housing Supply

To close the model and conduct counterfactual policy analyses, we specify the supply side of the housing market. The supply specification determines how changes in neighborhood demand induced by the disclosure policy are split between price adjustments and quantity responses. We use an iso-elastic neighborhood-level housing supply function, anchored at the observed equilibrium:

$$\log s_{jt}^{S}(P) = \log s_{jt}^{\text{obs}} + \eta_j \left(\log P - \log P_{jt}^{\text{obs}} \right), \tag{11}$$

where $s_{jt}^{S}(P)$ is the share of the overall market supplied by neighborhood j at price P, s_{jt}^{obs} and P_{jt}^{obs} are the observed share and price, and η_j is the neighborhood's housing supply elasticity. Anchoring the supply curve at the observed equilibrium means we do not need to estimate the level of supply: absent any change in demand, the model reproduces the observed prices and quantities.

In a counterfactual with beliefs $\mathbf{r}_t^{CF}$, equilibrium prices $\mathbf{P}_t^{CF}$ clear the housing market in every neighborhood:

$$s_{jt}^{S}\left(P_{jt}^{CF}\right) = s_{jt}^{D}\left(\mathbf{P}_t^{CF}, \mathbf{r}_t^{CF}\right) \quad \forall j, t. \tag{12}$$

where the bold letters represent vectors of prices and beliefs across all neighborhoods in year t. Because demand for each neighborhood depends on prices and beliefs everywhere, Equation (12) is a system of equations solved jointly within each year. We hold the overall market size fixed, so the outside option absorbs any change in total home purchases in California. We assume that the disclosure policy shifts housing demand but not the supply curve itself.

This specification nests the two supply scenarios we consider. In the short run, supply is perfectly inelastic ($\eta_j = 0$). Housing is durable, and in California new construction is constrained by permitting and land scarcity. With perfectly inelastic supply, demand shifts are fully capitalized into neighborhood prices, with no quantity response, although the composition of buyers within each neighborhood can still change. This scenario is most appropriate for studying the short-run price and sorting effects of the disclosure policy.

In the longer run, we allow quantities to respond: price increases in a neighborhood attract new construction, and price declines reduce it. We calibrate η_j using tract-level estimates of housing-unit supply elasticities from Baum-Snow and Han (2024), aggregated to neighborhoods as described in Section 4.2.6.

These elasticities are estimated over a decade-long horizon, so we interpret the resulting counterfactuals as comparisons of longer-run equilibria. When supply is elastic, declines in demand yield fewer home sales in addition to lower prices.

4 Data

We study the housing market in California from 2017-2023. This seven-year time span gives us four years of pre-policy and three years of post-policy coverage.

Estimating the demand model requires two primary data inputs, which we describe in turn. First, we construct a repeated cross section of the neighborhood choices and characteristics of single family home buyers in the state of California for our sample period. We also build a panel of "product characteristics" for each neighborhood that comprises the choice set for households in the model.

For the purposes of bringing the model to data, we define neighborhoods as the geographic intersection of Census Zip Code Tabulation Areas (ZCTAs) with the area subject to the risk disclosure. The state of California is divided into about 1,700 ZCTAs.[5] After taking the spatial intersection of the ZCTAs with the area subject to the disclosure policy, we are left with about 2,300 neighborhoods.

4.1 Household Choices and Characteristics

We construct a dataset of household's neighborhood choices and characteristics from a repeated cross section of home transactions derived from the CoreLogic OwnerTransfer data file. Our sample captures all arms-length transactions of single family homes in the state of California from 2017-2023.

4.1.1 Neighborhood Choice

We code household neighborhood choice by geocoding each transaction to our neighborhood definitions using parcel-level latitude and longitude from the CoreLogic data. By geocoding to a fixed GIS representation of the neighborhoods, we ensure that neighborhood choice outcomes are measured consistently across

[5]The set of ZCTAs do not fully cover the state of California - there are swathes of largely uninhabited Federal land that are not covered by any ZCTA. We implicitly exclude these areas from households' choice set.

all years in our dataset, even though ZCTAs were updated by the Census Bureau for the 2020 Census. We use the 2010 ZCTAs for the entire sample period.

4.1.2 Buyer Type Information

Households vary in an observable way along two dimensions: cash/mortgage financing status, and intended usage (primary vs. non-primary property usage). Both buyer financing and intended primary home usage are directly observable in the CoreLogic data. We define four observable buyer types as the intersection of these two binary characteristics. Table 1 shows the distribution of the four different buyer types in the data. The modal buyer is mortgage-financed and intends to purchase the home for primary use, so we set these buyers as the baseline or left-out category in the model. Overall, about 76% of buyers in our sample purchase a home with mortgage financing, and 64% of home purchases are intended as the buyer's primary residence. We cannot distinguish between different secondary uses, such as second homes, short-term rentals or long-term rentals, so we choose to focus on the primary vs. non-primary margin.

Table 1: Buyer Type Distribution

Financing	Purchase Type	Frequency
Mortgage	Primary	0.57
Mortgage	Non-Primary	0.19
Cash	Primary	0.07
Cash	Non-Primary	0.16
N		3,343,435

Notes. This table reports the distribution of buyer types in the CoreLogic transaction data, covering arms-length sales of single-family homes in California from 2017 to 2023, the demand estimation sample. The sample is restricted to purchases in the neighborhood-years used in estimation; N is the number of transactions. Buyers are categorized by financing status (mortgage or cash) and intended use (primary residence or non-primary, e.g., a second home or investment property). Most transactions are mortgage-financed purchases of primary residences. Cash buyers are disproportionately represented among non-primary purchases: cash finances 46% of non-primary purchases but 12% of primary-residence purchases. Frequencies may not sum to one because of rounding.

4.2 Neighborhood Characteristics

The other primary data input for the model is a panel of neighborhood characteristics. We construct a yearly panel that captures the "product characteristics" of each neighborhood in our dataset for each year from 2017 through 2023. For each neighborhood-year, we measure several characteristics: a home

price index P_{jt}, the annual burn probability R_j, realized wildfire events F_{jt}, market share s_{jt}, disclosure status D_j, a measure of insurance market tightness, and two measures of home insurance prices. The home price index, market share, disclosure status, and insurance conditions all vary at the neighborhood-year level, while wildfire risk is treated as time-invariant given that the underlying physical hazard does not change materially over our sample period. Table 2 provides summary statistics for the key neighborhood characteristics, for the entire sample as well as comparing neighborhoods subject to disclosure ($D = 1$) and those not subject to risk disclosure ($D = 0$). We describe the source and construction of each of these measures in turn.

4.2.1 Home Price Indices

Estimating the model requires neighborhood-level price indices that vary at the yearly level. We construct our own price indices using the universe of single family home transactions in California from CoreLogic. We follow Sieg et al. (2002) and estimate neighborhood price indices as neighborhood-year fixed effects in a hedonic regression of the following form:

$$\ln p_{kt} = a_{j(k),t} + H_k\Gamma + u_{kt}$$

Here, p_{kt} is the observed transaction price for house k sold in neighborhood $j(k)$ in year t. H_k is a vector of house characteristics that includes lot size, home square footage, the number of bedrooms and bathrooms, and more.

We link the sample of transactions from the CoreLogic OwnerTransfer file with detailed structure and property characteristics from the CoreLogic Property file to create our estimation sample, which spans about 3.8 million transactions over our eight year sample period.

To more flexibly estimate the hedonic model, we generalize the linear model above into the following form:

$$\ln p_{kt} = a_{j(k),t} + f(H_k) + e_{kt}$$

To estimate the model, we proceed in two steps, applying the logic of the Frisch-Waugh-Lovell theorem to

Table 2: Summary Statistics: Neighborhood Characteristics

	Overall	$D_j = 1$	$D_j = 0$
Panel A: Prices and market shares			
Log price index	13.2	13.1	13.2
	(0.501)	(0.567)	(0.447)
Price level ($1,000s)	595	605	587
	(642)	(924)	(314)
Market share ($\times 10^{-3}$)	0.395	0.169	0.554
	(0.485)	(0.248)	(0.545)
Panel B: Wildfire risk			
Burn probability R_j (%)	0.288	0.626	0.0482
	(0.598)	(0.802)	(0.129)
Cumulative fire exposure F_{jt} (%)	1.61	3.52	0.254
	(7.57)	(11.0)	(2.66)
Any prior fire (%)	38.4	59.5	23.5
	(48.6)	(49.1)	(42.4)
Panel C: Insurance market conditions			
Private premium ($/policy-year)	1,420	1,578	1,308
	(885)	(811)	(918)
FAIR premium ($/policy-year)	1,515	1,933	1,200
	(1,250)	(1,267)	(1,140)
FAIR share of policies (%)	4.33	7.09	2.38
	(9.34)	(12.0)	(6.18)
Panel D: Housing supply			
Supply elasticity η_j	0.457	0.592	0.361
	(0.285)	(0.225)	(0.285)
Housing units with imputed η (%)	17.1	27.7	9.60
	(37.4)	(44.2)	(29.4)
Neighborhood-years	15,579	6,461	9,118
Neighborhoods	2,339	994	1,345
ZIP codes	1,713		

Notes. Means with standard deviations in parentheses. Unit of observation is the neighborhood-year, where a neighborhood is the intersection of a ZCTA with the FHSZ-regulated area; $D_j = 1$ denotes the regulated portion, subject to the disclosure requirement from 2021. The price index P_{jt} is the ZCTA $\times$ FHSZ $\times$ year fixed effect from the hedonic regression, re-centered at the mean log sale price. Market share is the neighborhood's share of the state-year market for owner-occupied homes, adjusted for net owner-household migration out of California. Burn probability R_j is the annual burn probability from the USFS *Wildfire Risk to Communities* data (Scott et al., 2024), averaged over the neighborhood with housing-unit weights. Cumulative fire exposure F_{jt} sums the annual housing-weighted share of the neighborhood reached by wildfire over fire years 2012 to $t-1$, and any prior fire indicates $F_{jt} > 0$. Insurance variables are earned premium per unit of earned exposure from the CDI Community Service Statements, which end in 2022 and are reported at ZIP level only. The FAIR premium is observed for 91.8% of neighborhood-years, and is missing where FAIR exposure is zero. The supply elasticity η_j is the housing-unit supply elasticity of Baum-Snow and Han (2024), averaged over 2000 Census tracts with housing-unit weights; for tracts outside their sample it is imputed from tract observables, and neighborhood values are floored at zero. Imputed share is the percentage of the neighborhood's housing units in imputed tracts. Supply elasticities are missing for the 2 neighborhoods with no housing units.

first flexibly estimate the effects of the structural characteristics H_k on prices p_{kt} using a random forest model (Wright and Ziegler, 2017) to estimate the hedonic function $f(\cdot)$.

$$\ln p_{kt} = f(H_k) + u_{kt}$$

Then, having estimated the function $\hat{f}(\cdot)$ via random forest, we construct the first-step residuals:

$$\hat{u}_{kt} = \ln p_{kt} - \hat{f}(H_k)$$

To recover the zip-year fixed effects, we then run the following regression:

$$\hat{u}_{kt} = a_{j(k),t} + e_{kt}$$

Out of the box, the random forest already substantially outperforms the linear model, and after tuning the model hyperparameters, model performance (as assessed by RMSE or R^2) increases further.

A further advantage of the random forest approach is that it neatly handles missing data in the predictors by treating missings as a distinct, qualitatively different category than non-missing predictors. This approach is both more flexible and introduces less estimation error than imputation-based methods of addressing missing predictor variables.

Figure 3 compares the estimated neighborhood price indices from the random forest model and the linear hedonic model. The estimates from the random forest model are less strongly associated with the mean neighborhood sales price (the x-axis variable), which suggests that the random forest model does a better job of disentangling the relationship between underlying neighborhood quality (as measured by the price index) and observable home characteristics.

4.2.2 Wildfire Risk and Realized Exposure

Wildfire risk in California is highly spatially concentrated. Our primary measure of wildfire risk is the annual burn probability from the U.S. Forest Service's *Wildfire Risk to Communities* data (Scott et al., 2024). We compare this continuous, forward-looking measure to two others: historical wildfire perimeters and the discrete Fire Hazard Severity Zones (FHSZs) developed by the California Department of Forestry

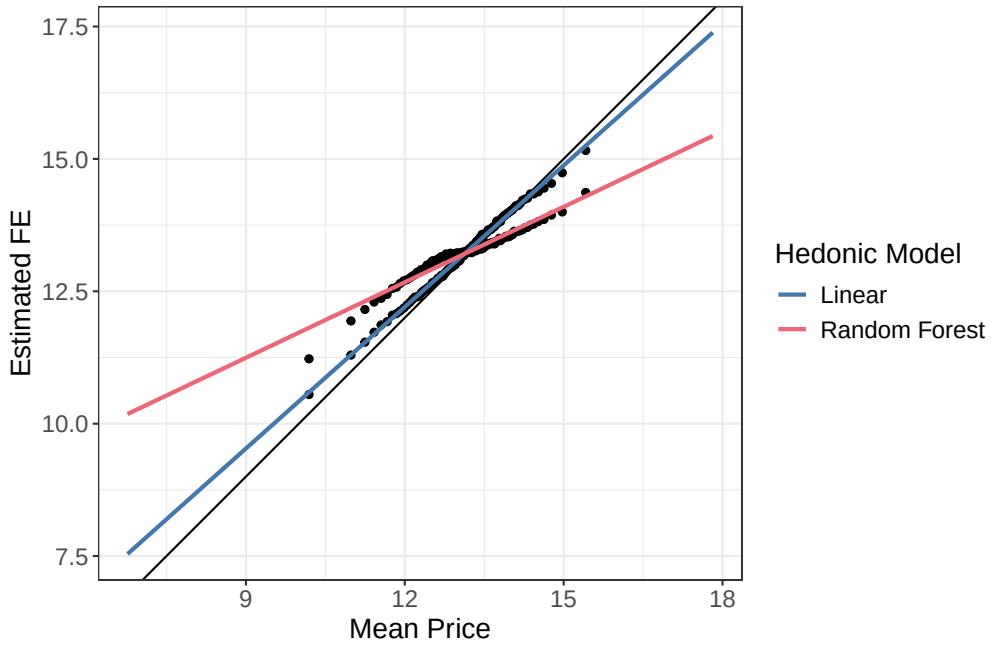
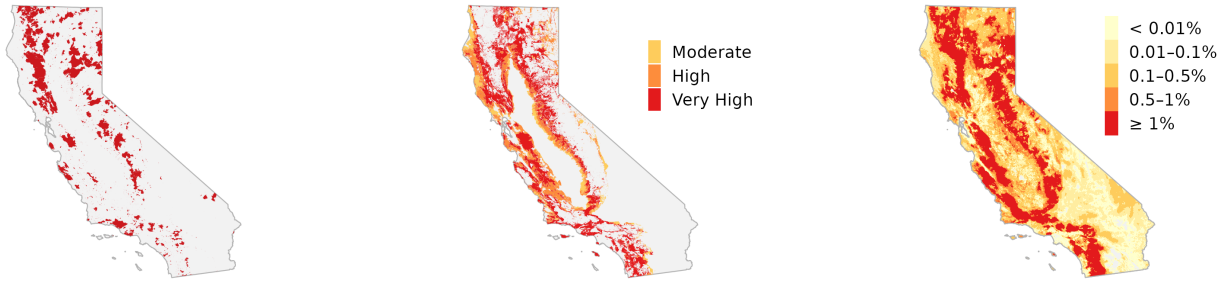


Figure 3: Comparison of Estimated Neighborhood Price Indices

Notes:
The black 45-degree line corresponds to mean observed neighborhood prices. The blue line shows price indices estimated with a linear hedonic model, while the red line shows estimates from a random forest model. Both approaches adjust for observable housing characteristics, but the random forest does so more flexibly. The greater divergence of the random forest estimates from the 45-degree line indicates that it more effectively removes the influence of compositional differences in housing stock (e.g., larger or higher-quality homes in higher-priced neighborhoods), yielding a cleaner measure of underlying neighborhood value.

and Fire Protection (Cal Fire). Figure 4 maps the three measures side by side.



(a) Wildfire Perimeters **(b)** Fire Hazard Severity Zones **(c)** USFS Burn Probability

Figure 4: Wildfire exposure and risk in California.

Notes:
Panel (a) shows wildfire perimeters from 2012 to 2023.
Panel (b) shows the Fire Hazard Severity Zones (FHSZs) used to define disclosure policy boundaries.
Panel (c) shows USFS annual burn probability, averaged to 1-km cells and grouped into classes on a log scale; areas with zero burn probability are unfilled.

All three measures identify similar high-risk regions at a broad geographic level. However, the FHSZ boundaries impose discrete cutoffs on what is fundamentally a continuous risk landscape, so nearby properties can face similar underlying risk but differ in disclosure status because of where the boundaries fall.

The Cal Fire FHSZ maps divide the state into four discrete fire risk categories: Low, Moderate, High, and Very High. In addition to serving as a publicly available source of fire risk information, the FHSZs were used by the state legislature to define the areas where sellers are legally required to disclose wildfire risk at the time of sale.

Burn probability is the annual probability that wildfire burns a given location. The Forest Service estimates it with the large fire simulation system that simulates many seasons of fire ignition and spread using data on fuels and vegetation, topography, and historical weather and ignition patterns. The simulations are run at a 270-meter resolution, then downscaled to 30 meters. In the downscaling, burn probability is spread from large patches of wildland into adjacent developed areas, which the fuels data otherwise classify as non-burnable. Homes at the wildland-urban interface receive a positive burn probability, while locations more than roughly a mile from a large patch of wildland have a burn probability of zero.

We aggregate burn probability to our ZCTA-disclosure neighborhoods using housing-unit weights. The Forest Service distributes 2020 Census housing units across building footprints on the same 30-meter grid (Jaffe et al., 2024), and we define R_j as the housing-unit-weighted mean burn probability over the cells in neighborhood j. Our measure corresponds to the expected annual burn probability of a randomly chosen

home in a neighborhood. Weighting by housing units instead of area prevents R_j from being driven by uninhabited wildland within a neighborhood. About two-thirds of California homes have a burn probability of zero.[6] Using a continuous measure of wildfire risk allows us to capture the substantial heterogeneity in wildfire risk within the discrete FHSZs that determine which areas are subject to the disclosure policy.

We also construct a measure of realized wildfire exposure at the neighborhood-year level, built to be consistent with burn probability. We use annual fire perimeters from the Cal Fire Fire and Resource Assessment Program (FRAP), keeping wildfires larger than one acre and excluding prescribed burns.[7] For each year, we treat the area inside the perimeters as burned and spread it into adjacent non-burnable land using the same procedure the Forest Service uses to spread burn probability into developed areas.[8] Annual exposure f_{jt} is the housing-unit-weighted mean of the resulting map over neighborhood j, the share of the neighborhood's homes reached by wildfire in year t. Because f_{jt} is built with the same housing weights and the same spreading procedure as R_j, the two are measured in the same units.

Cumulative exposure $F_{jt} = \sum_{s=2012}^{t-1} f_{js}$ sums annual exposure over fire years from 2012 through $t-1$. Each neighborhood therefore enters the estimation sample in 2017 with a five-year fire history, and exposure continues to accumulate over the sample period as fires occur. Fire exposure is heavily zero-inflated: 54% of our neighborhoods have no fire exposure from 2012-2022.[9]

4.2.3 Market Shares and Net Migration

We construct market shares from the same sample of arms-length purchases of single-family homes that forms the basis for our buyer sample. Let Y_{ijt} be an indicator equal to one if buyer i purchases a home in neighborhood j in year t. The market share of neighborhood j is

$$s_{jt} = \frac{1}{N_t} \sum_i Y_{ijt} = \frac{n_{jt}}{N_t},$$

[6] Neighborhoods with zero or near zero ($R_j < 10^{-6}$ burn probability make up 13% of our sample.

[7] The 1-acre threshold removes 8.9% of wildfire perimeters from 2012 to 2022, which together account for 91 of 13.6 million burned acres. Including them would change cumulative exposure F_{jt} in 2.9% of neighborhood-years, and would raise total cumulative exposure by less than 0.02%.

[8] The procedure applies three successive 510-meter circular moving averages on the 30-meter grid, spreading values only from patches of wildland of at least 500 hectares. We replicate the propagation using the Forest Service's published exposure-type layer, which was built with the same procedure.

[9] The information set of households in the model buying in market year t contains all fires up through year $t-1$, so buyers in 2023, the final year of our estimation sample, use fire exposure information through 2022.

where n_{jt} is the number of purchases in neighborhood j and the market size N_t is the number of transactions in our CoreLogic sample in year t plus the net number of owner households that move out of California, M_t. A neighborhood enters the choice set J_t in year t if it records at least two sales, the minimum needed to estimate its price index.

We measure M_t in households, the same unit as transactions, using the ACS 1-year PUMS. We count owner households whose householder moved from another state into California in the previous year, and owner households whose householder moved from California to another state, with tenure measured at the destination. M_t is the absolute value of the difference; the flow is out of California in every year of our sample. Measuring tenure at the destination counts the households that leave California and own a home elsewhere, the households choosing the outside option. Netting out in-movers avoids double counting, since in-movers who buy a home in California already appear among transactions.[10]

Net homeowner out-migration roughly doubled over the sample period, from 37,000 households in 2017 to about 100,000 in 2021 and 2022, before falling to 70,000 in 2023. Together with the decline in in-state sales in 2022 and 2023, the outside-option share $s_{0t} = M_t/N_t$ grows from 6.6% in 2017 to 19% in 2022 and 17% in 2023. Figure A2 plots net homeowner migration and the implied outside-option share over the estimation period.

4.2.4 Disclosure Status

California's wildfire risk disclosure requirement was signed into law in 2020 and took effect in 2021. The law requires sellers of residential properties in designated high-risk fire zones to provide buyers with a standardized written disclosure of the parcel's wildfire risk at the time of sale. The geographic scope of the requirement is defined by Cal Fire's Fire Hazard Severity Zone (FHSZ) maps: properties classified as High or Very High fire hazard severity are subject to mandatory disclosure.

We construct a neighborhood-level disclosure indicator $Disclosed_j$ that equals one for neighborhoods located within the disclosure area in years after the policy took effect (2021 onward), and zero otherwise. Neighborhoods that lie entirely outside the High and Very High FHSZs are never disclosed; neighborhoods that lie entirely within the disclosure area are disclosed from 2021 forward. Because we define neighbor-

[10]The ACS 1-year sample was not released for 2020. For 2020, we scale net domestic migration from the Census Population Estimates Program by the average ratio of net owner households to net domestic migration in 2019 and 2021.

hoods as the spatial intersection of ZCTAs with the FHSZ boundary, neighborhoods are either fully inside or fully outside the disclosure area by construction – a ZCTA that straddles the boundary is split into two separate neighborhoods, one disclosed and one undisclosed. This split adds about 600 neighborhoods to our choice set compared to a ZCTA-only neighborhood definition, and ensures that all neighborhoods in our sample have a single consistent treatment status.

4.2.5 Insurance Market Conditions

The market for homeowners insurance in California is facing pressures from changes in underlying risk of fires and strict regulation on insurance premiums. During our study period, several major insurers stopped writing new policies in the state, highlighting the strain in the homeowners insurance market.

We use data from the California Department of Insurance's Community Service Statement (CSS) to provide supplementary information about the status of the local homeowners insurance market. We use annual data from the CSS covering 2017-2022 to calculate annual ZIP code level coverage share of the California FAIR Plan, as well as two measures of insurance prices. The FAIR Plan is California's state insurer of last resort, which offers more expensive and less comprehensive coverage than private insurers. FAIR coverage excludes certain perils covered under standard homeowners' policies, such as theft, earthquakes, and liability coverage.[11] A higher share of FAIR Plan policies in a ZIP code signals that private insurance is scarce or prohibitively expensive.

The CSS provides detailed information on the number of policies and amount of coverage underwritten broken down by insurance provider, line of business, and ZIP code at an annual frequency. Our primary measure of insurance market tightness is the fraction of all homeowners policies underwritten in ZIP code j in year t that are written by the FAIR Plan:

$$\text{FAIR Share}_{jt} = \frac{\text{FAIR policies}}{\text{Total homeowners policies}}$$

The FAIR Plan coverage data serves as a proxy for the share of homeowners in a given ZIP code who have experienced difficulty obtaining insurance coverage through the private insurance marketplace. Figure 5 shows that reliance on the California FAIR Plan rose sharply between 2017 and 2022, but the increases

[11] Homeowners who obtain coverage under the FAIR Plan typically also require supplemental coverage for these excluded hazards from a private insurer.

were not uniform across the state. Average ZIP-level FAIR exposure in 2017 was 2.57%, which increased to 8.26% by 2022. Figure 5 shows the level of fair plan reliance in California in both 2017 and 2022.

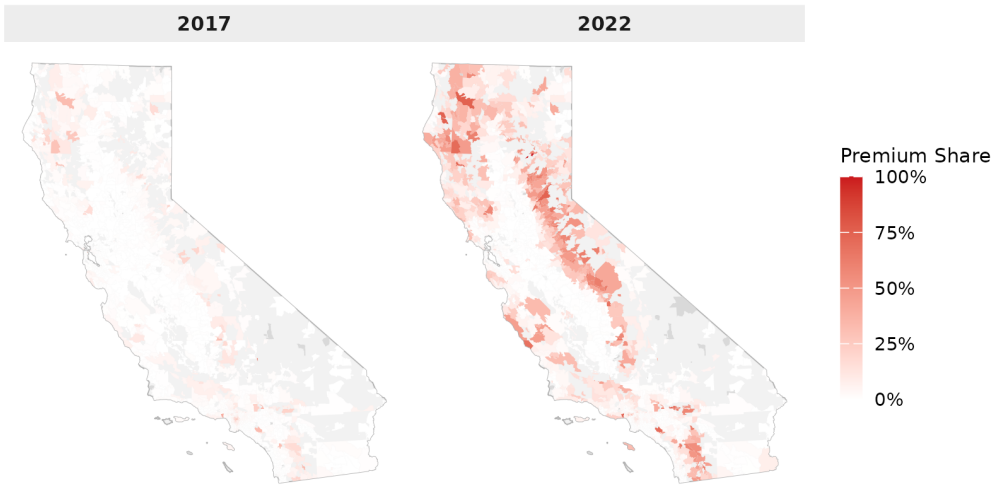


Figure 5: FAIR Plan Exposure in California

Notes:
Panel (a) 2017 share of total homeowners insurance premiums written by the California FAIR Plan.
Panel (b) 2022 share of total homeowners insurance premiums written by the California FAIR Plan.

Data come from the California Department of Insurance Community Service Statements. Between 2018 and 2020, the FAIR Plan's presence increased sharply, particularly in the Sierra foothills, along the North Coast, and the outskirts of San Diego. This shift reflects growing difficulties in securing standard homeowners insurance in wildfire-prone areas. The rising reliance on the FAIR Plan indicates increasing market withdrawal by private insurers and highlights the limited availability of affordable and comprehensive coverage for households facing high environmental risk.

The largest increases in FAIR Plan reliance were concentrated in the Sierra Nevada foothills, Northern California, and parts of Southern California—areas where wildfire activity intensified. This spatial pattern mirrors areas of intensified wildfire activity and reflects private insurers' withdrawal from high-risk markets, underscoring growing insurance scarcity in high-fire-risk areas of the state. As a result, we include insurance market conditions as a separate characteristic that households value in the model.

We also calculate two ZIP-level measures of insurance prices. Both measures capture *average* insurance prices in neighborhood j in year t for a given class of homeowners insurance policy. For both privately-underwritten and FAIR plan policies, we calculate the average premium as the total earned premiums for

that year divided by the total number of policy-years in that ZIP code for that year.

4.2.6 Housing Supply Elasticities

We take housing supply elasticities from Baum-Snow and Han (2024), who estimate the elasticity of the number of housing units with respect to house prices for individual census tracts, using changes in tract-level housing quantities and prices between 2000 and 2010. They identify local supply elasticities using commute-time weighted local labor demand shocks. We use their preferred quadratic finite-mixture specification. Because the estimates measure responses over a decade, we interpret them as longer-run elasticities, consistent with their role in our counterfactuals (Section 3.2). We use elasticities for housing units rather than floor space to match the margin on which market shares are measured.

We aggregate the tract elasticities to our neighborhoods using housing-unit weights. Each 30-meter cell on the Forest Service housing grid (Jaffe et al., 2024) takes the elasticity of the 2000 census tract that contains it, and η_j is the housing-unit-weighted mean over the cells in neighborhood j, the same weighting we use for burn probability. This assigns each neighborhood the supply elasticity faced by its typical home.

The estimates of Baum-Snow and Han (2024) cover metropolitan areas, which account for 6,782 of California's 7,049 tracts from 2000. The 267 uncovered tracts contain 4.5% of the state's housing units and are concentrated in rural, fire-prone areas. We impute elasticities for these tracts from a generalized additive model fit to the covered tracts, using tract observables that predict supply elasticity: housing density, the share of land with housing, distance to the nearest dense tract, land area, and burn probability.[12] Of the 2,337 neighborhoods with housing, 1,879 lie entirely in covered tracts, 408 entirely in imputed tracts, and 50 in both. Imputed tracts account for 6% of purchases overall but 18% of purchases in disclosed neighborhoods. Finally, we floor neighborhood elasticities at zero, which affects 19 neighborhoods with small negative estimates (the lowest of which is -0.09).

Supply elasticities vary widely across California (Figure 6). Dense coastal metropolitan areas are the most inelastic, with elasticities of roughly 0.15 to 0.2, while the Central Valley and the Sierra foothills range from 0.4 to 0.9. The mean neighborhood elasticity is 0.47, but the housing-unit-weighted mean is

[12]The model includes smooth functions of log housing units per square kilometer, the share of 30-meter land cells containing housing, log housing-weighted and log area-weighted burn probability, the share of land with positive burn probability, log land area, and log distance to the nearest tract with at least 1,000 housing units per square kilometer. In a leave-one-region-out validation, it predicts covered tracts with an R^2 of 0.67 and a root mean squared error of 0.13, against a standard deviation of 0.23, with no overall bias.

0.27, since most homes are in inelastic urban areas. Supply is more elastic where wildfire risk is higher: the correlation between η_j and $\log R_j$ across neighborhoods is 0.72, and disclosed neighborhoods average 0.60, compared with 0.37 for non-disclosed neighborhoods. The neighborhoods where disclosure lowers demand are therefore those where quantities, rather than prices, are most able to adjust.

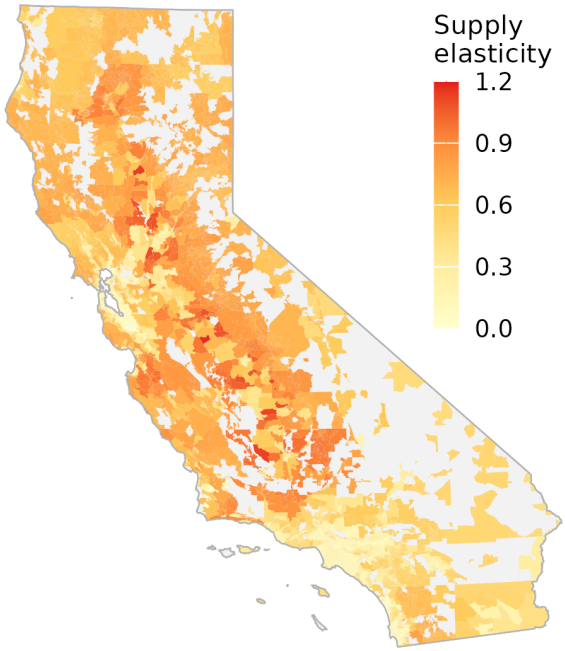


Figure 6: Housing Supply Elasticities

Notes. Neighborhood housing-unit supply elasticities η_j: tract-level elasticities from Baum-Snow and Han (2024), averaged over each neighborhood with housing-unit weights and floored at zero. Elasticities for tracts outside their sample are imputed from tract observables. Areas without housing units or outside any neighborhood are unfilled.

5 Estimation and Identification

Estimation of the demand model follows the procedures laid out in Berry et al. (1995) and Bayer and Timmins (2007) for estimating discrete-choice models of neighborhood demand. Because we observe individual buyers' neighborhood choices and types, we estimate the type-specific parameters by maximum likelihood on the microdata, rather than by the nested GMM approach used in settings where only aggregate market shares are observed.

It is convenient to separate the type-specific and common components of indirect utility:

$$V_{ijt} = \delta_{jt} + \theta^P_{k(i)} \log P_{jt} + \theta^R_{k(i)} E[r_j \mid F_{jt}, D_j] + \varepsilon_{ijt},$$
$$\delta_{jt} = \bar{\theta}^P \log P_{jt} + \bar{\theta}^R E[r_j \mid F_{jt}, D_j] + \xi_{jt},$$

where $k(i) \in \{0, 1, 2, 3\}$ is household i's buyer type. The coefficients θ^P_k and θ^R_k measure each type's price and risk sensitivity relative to the baseline type, mortgage-financed buyers of primary homes, for whom $\theta^P_0 = \theta^R_0 = 0$. The mean utility δ_{jt} captures the valuation of neighborhood j in year t common to all buyers: the baseline type's valuation of price and risk, and unobserved amenities ξ_{jt}.

This decomposition separates estimation into three steps. First, we estimate the disclosure rule parameters $\{\bar{r}, \tau\}$ by maximum likelihood on the cross-section of neighborhoods, outside the demand model. Second, we estimate the type-specific parameters $\left\{\theta^P_k, \theta^R_k\right\}$ and the belief precision κ by maximum likelihood, concentrating out the mean utilities δ_{jt} with the BLP contraction mapping. Third, we decompose the estimated mean utilities into the mean preferences $\bar{\theta}^P$ and $\bar{\theta}^R$ and unobserved amenities, using an instrumental variables regression to address the correlation between prices and amenities.

Our data are flows of buyers, while the demand model is static, although we segment the market by year t. In estimation, we take households' decision to purchase a home in year t as given, and use the choices of these buyers, conditional on purchasing, to identify the demand parameters. We leave unmodeled several dynamic features of the moving decision, such as selection into purchasing, housing endowments, and the continuation value of different neighborhoods. While this is a limitation, households' choices across neighborhoods still reveal much of their preferences for neighborhood characteristics. We discuss the identification of the type-specific and common components of utility in detail below.

5.1 Estimating the Disclosure Rule

Disclosure status is assigned at the neighborhood (ZCTA $\times$ disclosure status) level and does not vary over time, so we estimate the rule on the cross-section of neighborhoods $j = 1, \ldots, J$. Let $D_j \in \{0, 1\}$ indicate that neighborhood j lies in the FHSZ-regulated area, and let R_j denote its USFS burn probability.

The rule is a probit in log burn probability:

$$\Pr(D_j = 1 \mid R_j) = \Phi\left(\frac{\log R_j - \bar{r}}{\tau}\right), \tag{13}$$

where Φ is the standard normal CDF, $\bar{r}$ is the disclosure threshold on the log scale, and $\tau > 0$ governs how sharply the probability of disclosure rises around that threshold. Because the rule is defined in logs, we estimate it on the neighborhoods with positive burn probability.

We estimate $\{\bar{r}, \tau\}$ by maximum likelihood separately from the demand model. We treat the estimates as the commonly known disclosure rule that households take as given. In the belief model, households apply this rule to the unobserved burn probability r_j, interpreting a disclosure as a noisy signal that r_j lies above the threshold $e^{\bar{r}}$ (Section 3.1.3).

5.2 Maximum Likelihood Estimation

We leverage the individual-level data on actual neighborhood choices to estimate the parameters associated with the individual-specific component of utility μ_{ijt} via simulated maximum likelihood. This approach allows us to identify these parameters using a revealed-preference logic: the differential tendency of buyer types to choose neighborhoods with different bundles of characteristics reveals their preferences for those characteristics.

5.2.1 Identification

Because the mean utilities δ_{jt} absorb everything that is common to all buyers in a neighborhood-year, including unobserved quality ξ_{jt}, the parameters of the maximum likelihood step are identified from differences in where different types of buyers locate. The type-specific coefficients θ_k^P and θ_k^R measure each type's price and risk sensitivity relative to the baseline type, mortgage-financed primary-home buyers. They are identified from the systematic tendency of different buyer types to locate in neighborhoods with different price levels and beliefs about risk: if cash buyers and buyers of non-primary homes are less likely than mortgage-financed primary-home buyers to avoid neighborhoods they believe to be riskier, that differential sorting pattern identifies their risk coefficients, and buyer types that tend to purchase in more or less expensive neighborhoods are ascribed different price sensitivities.

The parameter κ is the precision of households' prior beliefs about neighborhood burn probability. It governs how strongly beliefs respond to new information: holding fixed the noise in the disclosure rule, τ, which we estimate separately, a lower κ means a more diffuse prior and larger belief updates in response to disclosure. Like the type-specific coefficients, κ is identified from the composition of buyer types across neighborhoods, through its effect on the beliefs that the θ_k^R multiply.

Identification of κ comes from comparing sorting patterns before and after the policy. Before 2021, households observe only the fire history, and in neighborhoods without recent fires beliefs are approximately proportional to R_j, so pre-period sorting across neighborhoods with different burn probabilities largely determines the θ_k^R. Once disclosure begins in 2021, beliefs in disclosed and non-disclosed neighborhoods diverge, by an amount determined by κ. Given θ_k^R, the model inverts the post-period shift in sorting to recover κ: if risk-sensitive buyer types move away from disclosed neighborhoods after 2021 by more than pre-period sorting on R_j alone would predict, the model rationalizes this with larger belief updates, and hence a lower κ. Realized fires provide additional variation, since κ also determines how much weight beliefs place on a neighborhood's fire history relative to R_j.

5.2.2 Estimation Procedure

Let $\Theta = \left\{\theta_k^P, \theta_k^R, \kappa\right\}$ collect the parameters estimated by maximum likelihood. Given a guess $\theta \in \Theta$, we invert the observed market shares s_{jt} to compute the mean utilities $\delta_t(\theta)$ that equate the model-implied market shares s_{jt}^D to the observed shares in each year, using the Berry et al. (1995) contraction mapping:

$$\delta_{jt}^{h+1} = \delta_{jt}^h + \log s_{jt} - \log s_{jt}^D\left(\delta_t^h; \theta\right),$$

where h indexes iterations. We start from $\delta_{jt}^0 = \log s_{jt} - \log s_{0t}$, the exact inversion when all household types have the same preferences,[13] and iterate until $\left\|\delta_t^{h+1} - \delta_t^h\right\| < \epsilon = \texttt{1e} - \texttt{10}$ in each year.

Given θ and $\delta_t(\theta)$, the probability that household i, of type $k(i)$, purchases in neighborhood j in year t is $\pi_{k(i)jt}$ from equation (9). We estimate θ on a random 5% sample of buyers in each year, 167,171 purchases

[13] During optimization, each inversion instead starts from the mean utilities at the previous parameter guess.

in total, by maximizing the concentrated log-likelihood

$$\ell(\theta) = \sum_t \sum_i \sum_{j \in J_t} Y_{ijt} \log \pi_{k(i)jt}\left(\theta, \delta_t(\theta)\right),$$

re-inverting the mean utilities at every evaluation of the likelihood. We maximize $\ell(\theta)$ with the L-BFGS quasi-Newton algorithm, with κ on a log scale to ensure non-negativity. We compute standard errors with the misspecification-robust sandwich estimator (Hansen, 2022, §10.15, §10.19), $\widehat{\mathcal{H}}^{-1}\widehat{\mathcal{I}}\,\widehat{\mathcal{H}}^{-1}$, where $\widehat{\mathcal{H}}$ is the Hessian of the concentrated log-likelihood and $\widehat{\mathcal{I}}$ is the outer product of the scores, clustered by neighborhood to allow for dependence among buyers in the same neighborhood across years. We compute both by finite differences, re-inverting the mean utilities at each perturbation. The standard error of κ is computed via the delta method.[14]

5.3 Estimating Mean Preferences

Decomposing the mean utilities δ_{jt} into their components presents an endogeneity challenge. We expect neighborhood prices P_{jt} to be correlated with unobserved amenities: desirable neighborhoods command higher prices precisely because of qualities we do not observe. Failing to account for this correlation biases the estimate of the baseline price sensitivity $\bar{\theta}^P$ toward zero. Risk beliefs may also be correlated with amenities, because natural beauty and access to mountains or forests are likely correlated with wildfire risk.

To recover the mean preferences of the baseline buyer type, mortgage-financed buyers of primary homes, we estimate

$$\delta_{jt} = \bar{\theta}^P \log P_{jt} + \bar{\theta}^R E[r_j \mid F_{jt}, D_j] + \varphi_j + \xi_{jt}, \tag{14}$$

where φ_j is a neighborhood fixed effect that absorbs time-invariant amenities and ξ_{jt} is the remaining time-varying component. We address the remaining correlation between P_{jt} and ξ_{jt} with an instrumental variable.

With neighborhood fixed effects, $\bar{\theta}^R$ is identified from within-neighborhood changes in beliefs. Because the burn probability R_j and the disclosure designation D_j are fixed over time, both are absorbed by φ_j. Identification of $\bar{\theta}^R$ therefore comes from the accumulation of fire exposure F_{jt} and from the introduction

[14] These standard errors treat the disclosure rule parameters $\bar{r}$ and τ as known.

of disclosure in 2021, which moves beliefs in disclosed and non-disclosed neighborhoods in opposite directions. We argue that these sources of variation are uncorrelated with changes in amenities ξ_{jt}, so that the estimate of $\bar{\theta}^R$ is consistent.

5.3.1 Price Instruments

To isolate variation in P_{jt} that is uncorrelated with ξ_{jt}, we adapt the shift-share instrument of Cook et al. (2023) to our context (Bartik, 1991; Goldsmith-Pinkham et al., 2020). The shifts are national growth rates in purchases by each buyer type k, and the shares are the propensities of each type to purchase in neighborhood j, measured from 2012 to 2016, before our sample period. The instrument is

$$W_{jt} = \frac{\sum_k g_{kt} N_k p_{jk}}{\sum_k N_k p_{jk}} = \sum_k \omega_{jk}\, g_{kt},$$

where g_{kt} is the national number of purchases by type-k buyers in year t relative to its 2012–2016 average, N_k is the number of type-k buyers in California in the base period, and p_{jk} is the share of buyers in that group who purchased in neighborhood j. Equivalently, W_{jt} averages the national growth rates with weights ω_{jk} equal to the base-period share of type-k buyers among neighborhood j's buyers. The first stage is

$$\log P_{jt} = \gamma_1 E[r_j \mid F_{jt}, D_j] + \gamma_2 W_{jt} + \varphi_j + u_{jt}.$$

The intuition behind the instrument is that neighborhoods historically popular with buyer types whose national presence is growing face rising demand, which, with inelastic housing supply, raises prices. The variation in W_{jt} comes from national shifts in buyer-type composition that are plausibly unrelated to changes in amenities in individual California neighborhoods.

The validity of the instrument rests on the exclusion restriction

$$E[W_{jt}\xi_{jt} \mid \varphi_j] = 0.$$

National changes in buyer-type composition must be uncorrelated with neighborhood-specific changes in amenities, conditional on the neighborhood fixed effects. The fixed effects relax this assumption by restricting identification to within-neighborhood variation in W_{jt} over time, which rules out correlation

between amenity levels and the base-period propensities p_{jk}. The main remaining threat is that national shifts in buyer-type composition coincide with changes in amenities over time. Because W_{jt} varies over time largely through the national growth rates, which are common to all neighborhoods, we do not include year fixed effects, which absorb most of the instrument's variation.

5.3.2 Standard Errors

We cluster standard errors in equation (14) by neighborhood, matching the maximum likelihood step and allowing for serial correlation in ξ_{jt} within neighborhoods. Because the mean utilities are inverted from market shares computed on the full population of transactions, they contain no sampling error of their own given the first-step parameters. Both $\hat{\delta}_{jt}$ and the beliefs depend on the first-step estimates, however: the mean utilities through the inversion, and the beliefs through $\hat{\kappa}$. Our second-step standard errors treat these first-step estimates as known and therefore do not reflect their sampling uncertainty.

6 Results

We now present and interpret the estimates of the demand model. Following the order estimation procedure, we first present the maximum likelihood estimates of the parameters associated with the individual-specific component of utility, and then the IV estimates of the mean value decomposition.

6.1 Disclosure Policy Estimates

Table 3 presents the maximum-likelihood estimates for the disclosure policy parameters for the probit model developed above. Column 1 shows the estimates from the standard probit, which imposes $\pi = 1$, while Column 2 allows the model to freely estimate π alongside the other parameters. The data strongly reject imposing $\pi = 1$, so our preferred specification is Column 2. Accounting for the ceiling probability shifts the disclosure threshold leftward (towards lower-risk neighborhoods) and halves the estimate of the noise parameter τ. Figure 7 plots the estimated disclosure policy rules over a binned scatterplot of the neighborhood risk scores R_j against the probability of disclosure $\Pr(D_j = 1)$.

Table 3: Disclosure Policy Parameter Estimates

	Estimate
$\bar{r}$ disclosure threshold, log scale	−7.38
	(0.0863)
τ probit scale	2.74
	(0.115)
$e^{\bar{r}}$ threshold burn probability (%)	0.0621
	(0.00536)
Log-likelihood	−1,034
McFadden pseudo-R^2	0.338
Neighborhoods	2,280
disclosed ($D_j = 1$)	993

Notes. Maximum likelihood estimates of the disclosure rule $\Pr(D_j = 1 \mid R_j) = \Phi\left((\log R_j - \bar{r})/\tau\right)$ on the cross-section of neighborhoods, where R_j is the housing-unit-weighted USFS annual burn probability. In the model, households apply the same rule to the unobserved burn probability r_j. $\bar{r}$ is the threshold on the log scale, τ governs how sharply the probability of disclosure rises around it, and $e^{\bar{r}}$ is the threshold expressed as an annual burn probability. Standard errors are from the inverse information matrix, by the delta method for $\bar{r}$, τ, and $e^{\bar{r}}$. Neighborhoods with zero burn probability, for which $\log R_j$ is undefined, and neighborhoods with no housing units are excluded.

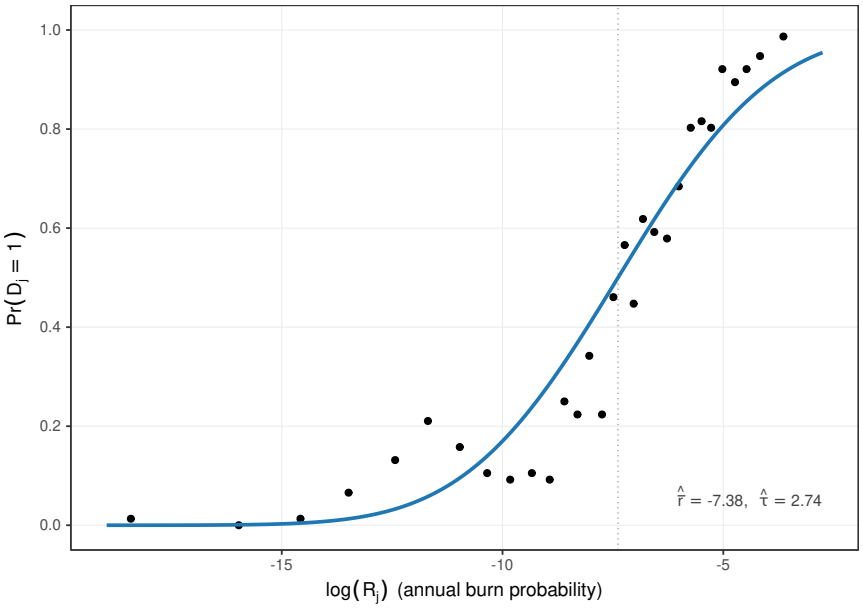


Figure 7: Estimated Disclosure Policy Rule

6.2 Demand Model Estimates

Table 4 presents the estimated parameters from the demand model using the two-stage estimation procedure described in section Section 5.

Panel A presents the estimates of risk belief sensitivity parameters θ_k^R, measured in percentage points

Table 4: Demand Model Estimates

	Estimate
Panel A: Risk sensitivity (per percentage point of burn probability)	
$\bar{\theta}^R$ baseline: Mortgage, Primary	-0.460
	(0.106)
θ_1^R Mortgage, Non-Primary deviation	0.749
	(0.0909)
θ_2^R Cash, Primary deviation	0.484
	(0.0658)
θ_3^R Cash, Non-Primary deviation	0.944
	(0.0754)
Panel B: Beliefs	
κ prior precision	35.0
	(8.26)
Panel C: Price sensitivity	
$\bar{\theta}^P$ baseline: Mortgage, Primary	-2.81
	(0.269)
θ_1^P Mortgage, Non-Primary deviation	-0.0101
	(0.0107)
θ_2^P Cash, Primary deviation	0.522
	(0.0657)
θ_3^P Cash, Non-Primary deviation	-0.0188
	(0.0119)
$\bar{\theta}^R$, 2017–2020 IV	-0.791
	(0.146)
Implied own-price elasticity, Mortgage, Primary	-2.81
range across types	$[-2.83, -2.29]$
First-stage F	92.9
Buyers (first step)	167,171
Neighborhood-years (second step)	15,506

Notes. Estimates of the demand model with Gamma–Poisson beliefs. The type-specific deviations θ_k^R and θ_k^P and the prior precision κ are estimated by maximum likelihood on a 5% sample of buyers, with mean utilities concentrated out; their standard errors are misspecification-robust sandwich estimates clustered by neighborhood (2,202 neighborhoods), and the standard error of κ is from the delta method. Deviations are relative to the baseline type, mortgage-financed buyers of primary homes (Mortgage, Primary). The baseline coefficients $\bar{\theta}^R$ and $\bar{\theta}^P$ are from an IV regression of the estimated mean utilities on beliefs and log price with neighborhood fixed effects, pooling 2017–2023, with log price instrumented by the shift-share instrument W_{jt}; standard errors are clustered by neighborhood. The memo row reports $\bar{\theta}^R$ from the same regression estimated on 2017–2020. Beliefs are measured in percentage points of annual burn probability, so $\bar{\theta}^R$ and θ_k^R are per percentage point. Price is measured in logs, so $\bar{\theta}^P + \theta_k^P$ is the own-price elasticity up to a factor $(1 - s_{jt}) > 0.99$. Standard errors treat the disclosure rule parameters $\bar{r}$ and τ as known, and the second-step standard errors treat the first-step estimates as known.

of annual burn probability. A positive estimate of the type-specific deviations θ_k^R implies that group k has a lower sensitivity to risk belief than buyers in the baseline category (mortgaged buyers of primary homes). We find a clear cut pattern in the type-specific risk sensitivities. Cash buyers are always less risk sensitive than mortgaged buyers, and buyers of primary homes are always more risk sensitive than buyers of secondary homes. The fact that $\theta_k^R > 0 \; \forall k$ suggests that mortgaged buyers of primary homes are the most risk sensitive group. The next most risk-sensitive group is mortgaged buyers of secondary homes. This pattern suggests that financing status (mortgage vs. cash) is a stronger determinant of buyer risk attitudes than primary vs. non-primary home usage.

To benchmark the scale of the risk sensitivity parameters θ_k^R, we convert the estimates into the willingness of different buyer types to pay to avoid a given increase in perceived wildfire risk.[15] Evaluating at the average home value in the sample of about \$595,000, types 1, 2, and 3 are respectively willing to pay about \$16,000, \$10,000, and \$20,000 *less* than the baseline type, mortgage-financed buyers of primary homes, to avoid a 0.1 percentage point increase in the annual probability that their home burns. For comparison, the average belief in our sample is 0.34 percentage points.

Panel B presents the model estimate of κ, the precision of the prior belief. κ can be interpreted as the weight that buyers place on the USFS burn probabilities R_j, measured in years of wildfire history. Higher values of κ correspond to more confidence in the risk map. Together with τ, the noise in the disclosure policy rule, this parameter governs the extent to which households update their beliefs in response to the disclosure signal D_j in the post-policy period. $\hat{\kappa} = 35.0$ implies that buyers weight their priors R_j from the USFS measure equivalently to 35 years of realized fire exposure. Because the base rates are so low, priors are still quite diffuse, and households still put a substantial weight on the disclosure signal, updating their beliefs substantially in response. Figure 8 shows the change in the distribution of model-estimated beliefs across neighborhood disclosure types between 2020 (the past pre-policy year) and 2021 (the first post-policy year).

Panel B reports the prior precision κ, which measures the weight buyers place on the USFS burn probability R_j in units of years of fire history. Higher values of κ correspond to more confidence in the risk map. The estimate $\hat{\kappa} = 35.0$ implies that buyers weight R_j as heavily as 35 years of realized fire exposure.

[15] Because price enters utility in logs and beliefs in percentage points, type k's willingness to pay to avoid a one-percentage-point increase in believed annual burn probability is approximately $-(\bar{\theta}^R + \theta_k^R)/(\bar{\theta}^P + \theta_k^P)$ as a share of home value. We report the difference from the baseline type.

By 2021, the nine years of observed fire history receive about 20 percent of the weight. However, because burn probabilities are so low, priors remain diffuse, and together with the noise in the disclosure rule, τ, this makes beliefs respond strongly to the disclosure signal D_j. Figure 8 shows how the distribution of beliefs in disclosed and non-disclosed neighborhoods changes between 2020, the last year before the policy, and 2021, the first year after.

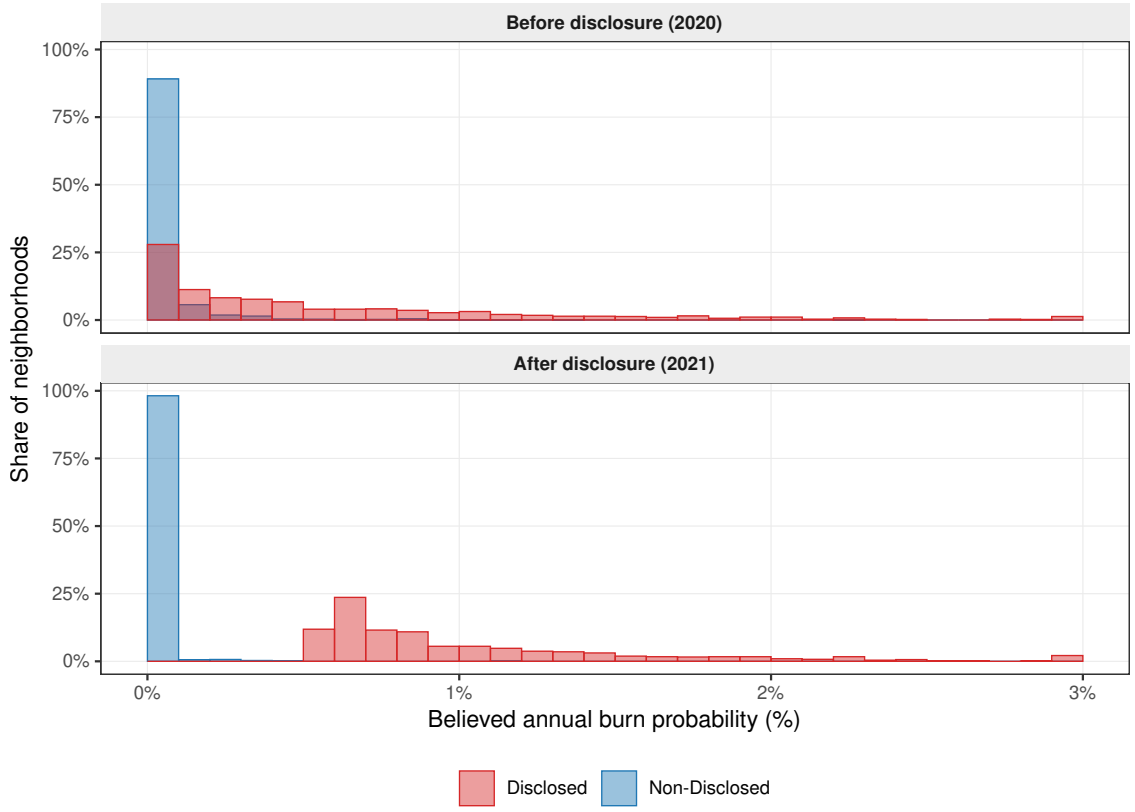


Figure 8: Belief Updating Across Neighborhood Types

Notes. Distribution of model-implied beliefs about the annual burn probability, $E[r_j \mid F_{jt}, D_j]$, across neighborhoods in the estimation sample, evaluated at the estimated parameters, separately for disclosed ($D_j = 1$) and non-disclosed ($D_j = 0$) neighborhoods. The upper panel shows beliefs in 2020, before disclosure, when beliefs are the fire-updated prior mean μ_{jt}; the lower panel shows beliefs in 2021, after households observe D_j. Beliefs in 2021 also incorporate fire exposure during 2020. Bars are 0.1 percentage points wide and show the share of each group's neighborhoods; beliefs above 3% (10 disclosed neighborhoods in 2020 and 17 in 2021) are included in the last bar. The sample contains 2,230 neighborhoods in 2020 and 2,248 in 2021.

The disclosure policy has two effects on beliefs. First, it drives a large divergence in beliefs between disclosed neighborhoods, where model-implied beliefs rise sharply, and non-disclosed neighborhoods, where they fall slightly. In 2021, the average gap in believed annual burn probability between disclosed and non-disclosed neighborhoods is 1.03 percentage points, compared with 0.55 percentage points had households updated on the 2020 fires alone, an increase of 0.48 percentage points. Second, disclosure

reduces the dispersion of beliefs within each group: relative to beliefs without disclosure, the standard deviation of beliefs falls by 16% among disclosed neighborhoods and by 66% among non-disclosed neighborhoods.

Panel C of the table reports the estimated price sensitivity parameters θ_k^P. The mean price sensitivity is about -2.81, which corresponds to the neighborhood-level own price elasticity up to a factor of $1 - s_{jt} > 0.99$ due to the small neighborhood-level market shares. This estimate comes from the IV decomposition of the mean values δ_{jt}.[16] The IV is well-powered (first-stage $F = 92.9$), and increases the magnitude of the estimated price sensitivity substantially compared to the OLS estimate. This is the expected direction of bias resulting from a correlation between prices P_{jt} and amenities ξ_{jt}. The price sensitivity parameters θ_k^P capture the difference price sensitivity relative to the left-out category, mortgaged buyers of primary homes. We estimate that cash buyers of primary homes are about 19% less price sensitive than the mortgaged buyers of primary homes; neither of the other two types has an economically meaningful or statistically significant estimated price sensitivity deviation. The positive price deviation for cash buyers of primary homes reflects a consistent pattern in the raw data: in every year from 2017 to 2023, the neighborhoods they choose have a price index 2.9 to 4.5 log points higher, on average, than those chosen by mortgage-financed buyers of primary homes

Interestingly, although the cash buyers of non-primary homes consistently buy in neighborhoods with lower price indices, the model does not rationalize this with a strong negative price deviation. Instead, because the neighborhoods they choose tend to also be quite risky, the model ascribes most of their observed sorting behavior to sorting on risk rather than sorting on price.

7 Counterfactual Exercises

We use the demand estimates from Section 6 and the supply model from Section 3.2 to evaluate the effects of the disclosure policy. In each exercise, we hold the estimated preferences and unobserved amenities ξ_{jt} fixed, change the information available to households, and solve for the new equilibrium prices and choices. We first simulate the housing market without the disclosure policy, so that beliefs evolve only with the fire history, and measure the policy's effects as the difference between the observed

[16] The full set of OLS, first stage, and IV regression estimates are reported in Table B1.

equilibrium and this counterfactual. We consider both short-run responses, with perfectly inelastic housing supply, and longer-run responses, using the supply elasticities from Baum-Snow and Han (2024).

We then evaluate alternative designs of the disclosure policy. First, we study the role of noise in the disclosure rule by setting $\tau = 0$, so that the designation perfectly reveals whether a neighborhood's burn probability lies above the threshold. Second, we consider a different way to increase the information content of disclosure: replacing the binary designation with a score from 1 to 10 based on deciles of neighborhood burn probability.

Finally, we ask whether inelastic housing supply limits the reallocation of risk induced by risk disclosure. Supply is least elastic in low-risk coastal metropolitan areas, so much of the demand that disclosure shifts toward safer neighborhoods may raise prices there rather than add homes. To assess this, we simulate the effect of the policy with more elastic supply in lower-risk areas.

7.1 Policy Evaluation: No Disclosure

We evaluate the effects of the disclosure policy on household sorting, equilibrium prices, and the incidence of wildfire risk by comparing the observed equilibrium with a counterfactual in which the disclosure policy is turned off in 2021–2023. In the counterfactual, beliefs continue to evolve with the fire history but do not respond to disclosure; we hold the estimated preferences and unobserved amenities fixed and solve for the new equilibrium prices and choices.

Table 5 presents the effects of the policy under short-run (fixed) and longer-run (elastic) housing supply. Disclosure widens the gap in beliefs between disclosed and non-disclosed neighborhoods by 0.47 percentage points of annual burn probability. Nearly all of this change comes from disclosed neighborhoods, where beliefs rise by 0.45 percentage points; beliefs in non-disclosed neighborhoods fall by only 0.02 percentage points, because they were already near zero. Prices fall in disclosed neighborhoods relative to non-disclosed neighborhoods, by 0.79% with fixed supply and 0.69% with elastic supply. With elastic supply, part of the decline in demand materializes as fewer sales in disclosed neighborhoods (0.22%) rather than lower prices, but the supply assumption otherwise changes little. In both scenarios, 2.25% of households choose different neighborhoods due to the policy. The share of households leaving the state is essentially unchanged, although mortgage-financed buyers of primary homes become slightly more likely

to leave (Appendix Table B2).

These price effects are much smaller than the roughly 4% decline estimated at the disclosure boundary by Bass and Fogarty (2025), but close to the roughly 1% reduction found when flood risk scores were randomly shown to Redfin users (Fairweather et al., 2023), a setting that also isolates the effect of information. Prices fall little because disclosure raises demand for disclosed neighborhoods among buyers of non-primary homes, which largely offsets the decline among the most risk-averse buyers: the policy changes who buys more than what they pay. The price effect is accordingly sensitive to $\bar{\theta}^R$, reaching 6.4% under the pre-period estimate (Appendix Table B3), and a boundary comparison also captures channels our model excludes, such as defensible-space requirements, building standards, and insurance. Effects vary across neighborhoods, with the largest declines along the coast and small increases in about a third of disclosed neighborhoods (Appendix Figure A3).

Figure 9 shows how the policy changes the sorting of buyer types under fixed supply. Panel (a) shows the composition of buyers in disclosed neighborhoods. The share of buyers in disclosed neighborhoods who are mortgage-financed primary-home buyers falls by 7.4 percentage points, while the shares of mortgage-financed and cash buyers of non-primary homes rise by 3.0 and 4.2 percentage points. The share of cash buyers of primary homes rises by only 0.2 percentage points: they are less risk averse than the baseline type, so they too shift toward disclosed neighborhoods relative to it, but less than buyers of non-primary homes, who take most of the places that mortgage-financed primary-home buyers give up. Because the composition of buyers depends on the differences in risk sensitivity across types, which are identified from cross-sectional sorting, these sorting responses are nearly identical under the pre-period estimate of $\bar{\theta}^R$, even though the price effects are much larger.

Panel (b) of Figure 9 shows how this re-sorting changes exposure to wildfire risk. The average burn probability of the neighborhoods chosen by mortgage-financed primary-home buyers falls by 9.4%, while it rises by 5.3% and 8.5% for mortgage-financed and cash buyers of non-primary homes; for cash buyers of primary homes it is essentially unchanged. As a result, the gap in exposure between the most risk-sensitive buyers (mortgage-financed buyers of primary homes) and the least risk-sensitive buyers (cash buyers of non-primary homes) widens by about 25% under both supply assumptions. The reallocation is similar when exposure is measured by fires that actually occurred rather than by burn probabilities: the gap in realized fire exposure from the year of purchase through 2023 widens by 29%. These realized fires are outside of

Table 5: Counterfactual Policy Effects

	Fixed supply ($\eta_j = 0$)	Elastic supply (η_j, Baum-Snow–Han)
Panel A: Beliefs (change, percentage points)		
Disclosed ($D_j = 1$)	0.450	
Not disclosed ($D_j = 0$)	−0.017	
Panel B: Prices (% of value)		
Disclosed ($D_j = 1$)	−0.65	−0.56
Not disclosed ($D_j = 0$)	0.14	0.13
Difference	−0.79	−0.69
Panel C: Exposure gap between types (% change)		
Burn probability	24.9	25.3
Realized fire exposure	29.1	29.5
Panel D: Households who choose differently (%)		
All households	2.25	2.25

Notes. Effects of the disclosure policy: the observed equilibrium minus the counterfactual without disclosure, in which beliefs evolve only with the fire history, pooled over 2021–2023 with years weighted by market size. Estimated preferences and unobserved amenities are held fixed, with $\bar{\theta}^R = -0.460$ and $\bar{\theta}^P = -2.81$ from the pooled IV. Fixed supply holds each neighborhood's sales at their observed level; elastic supply uses the housing-unit supply elasticities of Baum-Snow and Han (2024), with supply curves passing through the observed equilibrium. Belief changes are sales-weighted means, in percentage points of annual burn probability, and do not depend on the supply assumption. Prices are sales-weighted log-point differences ($\times 100$); the difference row is the disclosed minus the not-disclosed effect. Panel C reports the percent change in the difference in exposure between Cash, Non-Primary and Mortgage, Primary buyers, the least and most risk-sensitive types, measured by the buyer-weighted mean burn probability of the neighborhoods each type chooses and by the mean annual housing-weighted fire exposure of those neighborhoods from the year of purchase through 2023; Figure 9 shows exposure for each type. Panel D is the minimum share of households, including those leaving California, who choose differently: the total variation distance between the two choice distributions, by type and weighted by type shares. The sales-weighted mean estimated elasticity is 0.34 (0.55 in disclosed and 0.29 in not-disclosed neighborhoods). Appendix Table B2 reports the full set of outcomes, including perfectly elastic supply.

the information set of buyers in the model at the time of purchase, and this pattern confirms the reallocation of risk based on the hazard map translates to real-world wildfires.

7.2 Alternative Information Policy Designs

We next evaluate how alternative designs of the disclosure policy would change sorting and the allocation of wildfire risk. We compare each design with the same counterfactual without disclosure used above; throughout, housing supply is fixed and we use the pooled estimate of $\bar{\theta}^R$. We consider two ways to change the information that disclosure conveys.

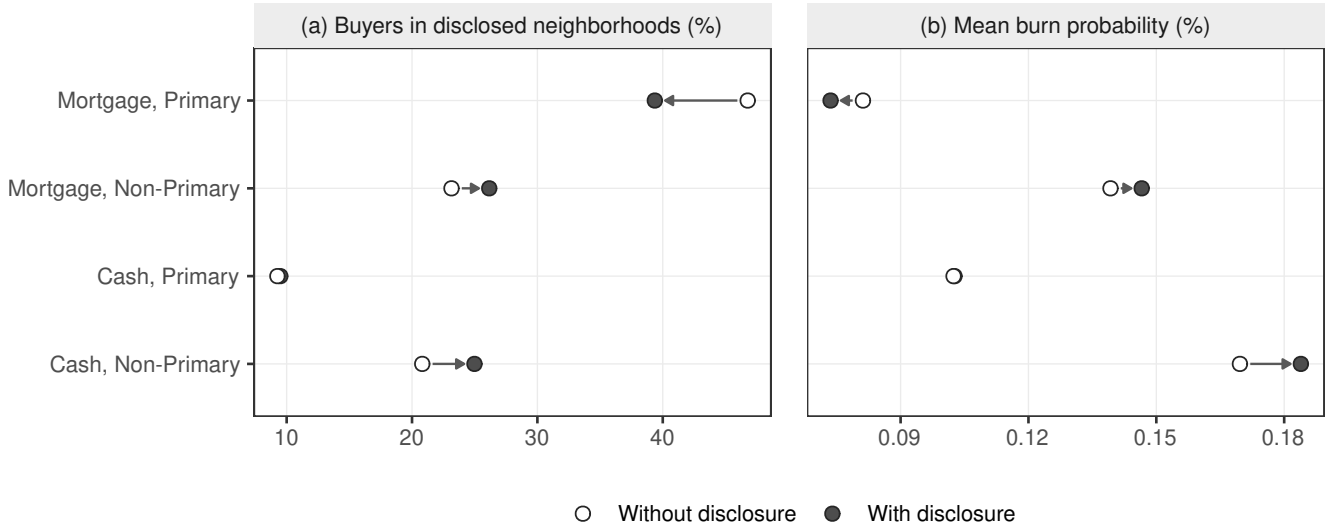


Figure 9: Changes in Sorting Behavior Under Counterfactual

Notes. Effects of the disclosure policy on where each buyer type purchases, comparing the observed equilibrium (filled) with the counterfactual without disclosure (hollow), in which beliefs evolve only with the fire history. Panel (a) shows each type's share of buyers in disclosed neighborhoods; panel (b) shows the mean USFS burn probability of the neighborhoods each type chooses, weighted by buyers. Values are pooled over 2021–2023, weighting years by market size. Estimated preferences and unobserved amenities are held fixed; housing supply is perfectly inelastic, so each neighborhood's sales are held at their observed level and only prices and the composition of buyers adjust.

The first keeps the binary designation but changes how precisely it reflects risk. We let the noise in the disclosure rule go to zero, $\tau \to 0$, so that households read the designation as revealing whether a neighborhood's burn probability lies above the threshold. We consider two versions. In the first, we keep the observed designations, which isolates how households' interpretation of the label matters, holding the labels themselves fixed. Because the observed designations do not follow the hazard map exactly, reading them as precise can also mislead households about some neighborhoods. In the second, we reassign designations from the hazard map, designating neighborhoods with $R_j > e^{\bar{r}}$. We evaluate the map-based designations both with the noise of the estimated rule and without it, which separates the effect of relabeling neighborhoods from the effect of reading the labels precisely.

The second strategy replaces the binary designation with a score from 1 to 10, based on deciles of burn probability across California neighborhoods. This design resembles the 1–10 climate risk scores, including wildfire risk scores from First Street, that Zillow displayed on home listings from 2024 until late 2025, when it removed them after complaints from the California Regional Multiple Listing Service (Brown, 2025). We again consider a noisy version, in which households read the score with the noise of the estimated disclosure rule, and a precise version, in which the score reveals the range of burn probabilities

that contains a neighborhood's risk.

Table 6 presents the results. The first column, the noisy binary designation with the observed labels, corresponds to the actual policy in California. Panel A summarizes the information each design conveys: the share of the prior variance of risk that it resolves and the mean absolute change in beliefs across neighborhoods. Panel B reports the change in the average burn probability of the neighborhoods chosen by each buyer type, along with the change in the gap between the least and most risk-sensitive types, cash buyers of non-primary homes and mortgage-financed buyers of primary homes. Panel C reports the share of households who choose differently because of the policy. We summarize each design by the reallocation of risk it achieves and by the re-sorting it requires to achieve it. The model does not include moving or transaction costs, so we treat the share of households who choose differently as a measure of the disruption a design imposes rather than as a welfare cost. Appendix Table B4 reports beliefs, prices, and outside-option shares for each design.

The two measures of information in Panel A behave differently: resolving uncertainty about risk is not the same as moving beliefs. The precise decile score resolves the most prior variance yet changes beliefs the least, and, as we discuss below, a design can also move beliefs substantially in ways that do not track underlying risk.

The first important result is that making the binary designation precise (by letting $\tau \to 0$) makes the policy less effective on both dimensions. The widening of the exposure gap falls by about 3 percentage points with either the observed or the map-based designations, while the share of households who choose differently rises by 0.3 to 0.4 percentage points. When the rule is noisy, households weigh the designation against their prior beliefs and discount it where the two conflict. Without noise, beliefs become nearly uniform within each label: they rise most in the safest disclosed neighborhoods and fall most in the riskiest non-disclosed neighborhoods. Risk-averse buyers move out of disclosed neighborhoods that are relatively safe and into non-disclosed neighborhoods that are relatively risky, generating more re-sorting that reallocates less risk. The effect is strongest with the observed designations, under which some non-disclosed neighborhoods are high-risk according to the hazard map.

Making the signal finer by moving from two risk categories to ten with the decile scores improves the results along both dimensions. The noisy decile scores achieve roughly the same level of risk re-allocation with a third less re-sorting. Making the decile scores precise further improves the effects: the amount of

Table 6: Alternative Disclosure Policy Designs

	Binary label				Decile score	
	Observed		From map		From map	
	Noisy	Precise	Noisy	Precise	Noisy	Precise
Panel A: Information						
Prior variance resolved (%)	12.2	16.3	12.2	16.3	17.8	64.1
Mean absolute belief change (pp)	0.096	0.109	0.111	0.126	0.099	0.055
Panel B: Mean burn probability of buyers' neighborhoods (% change)						
Mortgage, Primary	−9.4	−8.5	−11.0	−10.5	−8.7	−11.6
Mortgage, Non-Primary	5.3	5.0	6.3	6.3	3.9	3.3
Cash, Primary	0.2	0.7	0.8	1.4	−1.0	−4.9
Cash, Non-Primary	8.5	7.2	9.4	8.3	8.9	16.2
Gap between extreme types	24.9	21.5	28.3	25.6	25.1	41.8
Panel C: Households who choose differently (%)						
All households	2.25	2.54	2.68	3.07	1.50	0.85

Notes. Effects of alternative disclosure designs: each design minus the counterfactual without disclosure, in which beliefs evolve only with the fire history, pooled over 2021–2023 with years weighted by market size. The binary label either keeps the observed designations (observed) or assigns them from the map (from map), designating neighborhoods with $R_j > e^{\bar{r}}$; the decile score assigns each neighborhood its decile of R_j among California neighborhoods. Households read a label either with the noise of the estimated disclosure rule (noisy) or as revealing the range that contains r_j (precise). The noisy binary label with observed designations is the actual policy. Supply is fixed, and estimated preferences and unobserved amenities are held fixed, with $\bar{\theta}^R = -0.460$ and $\bar{\theta}^P = -2.81$ from the pooled IV. Prior variance resolved is the share of the prior variance of r_j that the design resolves before households observe their labels, $\mathrm{Var}(E[r_j \mid S_j])/\mathrm{Var}(r_j)$, a property of the signal structure. Panel B reports the percent change in the buyer-weighted mean burn probability of the neighborhoods each type chooses; the gap between extreme types is the percent change in the difference between Cash, Non-Primary and Mortgage, Primary buyers, the least and most risk-sensitive types. Panel C is the minimum share of households, including those leaving California, who choose differently. Appendix Table B4 reports beliefs, prices, and outside-option shares.

risk re-allocation increases by 16 percentage points while the share of households choosing a different neighborhood falls by a further factor of almost 2. Moving to the finer-grained disclosure measure makes beliefs update relatively proportionately to the differences in underlying risk.

Making the signal finer by moving from two risk categories to ten improves the policy. The noisy decile score achieves roughly the same reallocation of risk as the actual policy with a third less re-sorting. With ten categories, beliefs can track differences in risk that the binary designation groups together, so fewer households move in response to the label alone. Making the score precise improves the policy further, widening the exposure gap by about 17 percentage points more than the noisy decile score while nearly halving the share of households who choose differently. The precise score mostly confirms the beliefs of households in lower-risk neighborhoods and concentrates the change in beliefs in the highest decile, where the open-ended top category raises beliefs sharply. Risk-averse buyers leave the highest-risk neighborhoods, and few other households need to move. Precision has opposite effects on the two designs: it further improves a score fine enough to separate neighborhoods by risk, but hurts a binary designation that groups very different neighborhoods together.

8 Discussion

This paper studies how climate risk information shapes housing markets through its effects on household beliefs and residential sorting. We develop a structural model of neighborhood choice in which households' beliefs about neighborhood wildfire risk respond to a risk disclosure policy, and we use it to trace how those belief changes move prices and reallocate buyers across neighborhoods. Disclosure has modest effects on prices but meaningful effects on the incidence of wildfire risk. The buyer composition in high-risk neighborhoods shifts toward the those least sensitive to that risk, cash buyers and buyers of non-primary homes, and away from mortgage-financed primary-residence buyers. This pattern is consistent with insurance and lending requirements that make high-risk homes costlier for mortgage-financed buyers, though the model does not separate these constraints from preferences. The design of the disclosure policy also matters. Counterfactual simulations show that the precision and form of the label govern how much risk is reallocated relative to the disruption it causes, measured by the share of households who choose differently.

Taken together, these results suggest that information interventions may do less to reduce aggregate exposure to environmental risk than to reallocate it across households. In the short run the housing stock in high-risk areas is fixed, so disclosure changes who bears the risk rather than how much risk is borne, and even with elastic supply aggregate exposure barely changes.

8.1 Sorting Responses to Wildfire Risk Information

Disclosure reallocates housing in high-risk neighborhoods across buyer types, and the model attributes this reallocation to differences in sensitivity to wildfire risk. The estimates match the reduced-form findings of Bass and Fogarty (2025), who document lower prices in disclosed neighborhoods alongside more cash purchases and fewer purchases of primary residences. In the model, disclosure raises believed risk in disclosed neighborhoods; mortgage-financed buyers of primary residences, the type most sensitive to that risk, shift away, and cash buyers and buyers of non-primary homes take their place. The types that move in are no more sensitive to price than those who leave, so the re-sorting is primarily driven preferences for fire risk.

This differs from sorting over flood risk in Bakkensen and Ma (2020), who find that lower-income and minority households sort into high-risk floodplains to take advantage of lower prices. There, the households who bear more risk are those most responsive to price; here, they are those least sensitive to risk. Part of the difference may reflect data: their mortgage records exclude cash purchases, which account for much of the reallocation in our setting.

The two patterns have different distributional consequences. The buyers who replace mortgage-financed primary-home buyers in disclosed neighborhoods are mostly buyers of non-primary homes. In California, mortgage-financed buyers of second homes and investment properties are older, have higher incomes, and are less likely to be Hispanic or Black than buyers of primary residences, and nationally, households that own such properties have several times the net worth of other homeowners. Cash buyers of non-primary homes purchase below local median prices and often resell within a year, consistent with investor behavior.[17] Disclosure therefore shifts wildfire risk toward households plausibly better able to bear it, rather than toward

[17] Demographics of mortgage-financed buyers are from HMDA records of California home-purchase loans, 2018–2023, by occupancy type; net worth is from the Survey of Consumer Finances, 2019–2025; and purchase prices and resales of cash buyers of non-primary homes are from our CoreLogic deed records.

households who accept risk because they cannot afford to avoid it.

8.2 Financing Constraints and the Role of Cash Buyers

Our results also highlight the importance of financing status as a key, yet underexplored, dimension of heterogeneity in housing market responses to climate risk. Much of the existing literature focuses on mortgage-financed transactions, often due to data availability, implicitly treating these buyers as representative of the broader market. However, our analysis shows that while cash buyers are a minority of transactions (24% in our sample), they play an important role in shaping housing markets in high-risk areas.

Cash buyers differ from mortgage-financed buyers along several economically meaningful dimensions. Because mortgage lending typically requires proof of homeowners insurance, mortgage-financed buyers face binding constraints when insurance becomes costly or unavailable in high-risk areas. In contrast, cash buyers are less directly constrained by insurance requirements and may be more willing to accept wildfire risk. Consistent with this mechanism, we find that cash buyers are less risk sensitive than mortgage-financed buyers, and after disclosure the exposure of cash buyers of non-primary homes, rises the most.

Although these buyers account for a minority of transactions, they are often the marginal participants in high-risk neighborhoods and therefore play a central role in determining equilibrium prices and neighborhood composition. Analyses that exclude non-mortgage transactions thus risk overlooking a key channel through which climate risk is reallocated across households. More generally, our findings suggest that incorporating financing into models of housing demand is essential for understanding how climate risk and related policies shape both market outcomes and the distribution of exposure.

8.3 The Design of Risk Information

Our counterfactuals show that how risk information is delivered matters as much as whether it is delivered. The actual policy conveys risk through a coarse binary designation, and its effects come from how buyers combine that designation with what they already believe. Making the designation more precise does not help: when households read the binary label as exact, beliefs become nearly uniform within each category, rising most in the safest designated neighborhoods and falling in the riskiest undesignated ones, so the policy induces more re-sorting while reallocating less risk. Finer information performs better. A

ten-category risk score achieves the same reallocation of risk as the actual policy with a third less re-sorting, and a precise score widens the gap in exposure between the most and least risk-sensitive buyers by 42 percent while fewer than one percent of households choose differently.

The most important dimension of a risk information policy is its ability to separate the truly high-risk neighborhoods from the rest. Burn probabilities are highly right-skewed, and most of the variation in risk that matters for sorting is concentrated in a small set of neighborhoods at the top of the distribution. The current binary designation groups those neighborhoods with many moderately risky ones, so buyers do not fully respond to the underlying risk. They under-update for the truly high-risk neighborhoods and over-update for the moderate risk neighborhoods grouped into the same category. A finer designation lets beliefs rise sharply where risk is highest while largely confirming what buyers already believe elsewhere. This concentrates re-sorting where it most effectively re-allocates risk exposure. These results are relevant to ongoing debates about risk disclosure, including the recent removal of property-level climate risk scores from some major listing platforms (Brown, 2025): from the perspective of reallocating risk with minimal disruption, scores that distinguish truly high-risk properties may be more valuable than coarse designations.

9 Conclusion

Climate change is reshaping the distribution of risk across households, but the allocation of that risk depends not only on underlying hazards, but on what households know when making location decisions. This paper studies how information about wildfire risk affects housing market outcomes by developing and estimating a structural model of residential sorting in which households beliefs about underlying wildfire risk update in response to wildfire events and a risk disclosure policy. Exploiting the introduction of California's wildfire risk disclosure policy, we estimate that beliefs about risk in disclosed and non-disclosed neighborhoods diverge substantially, changing the composition of buyers, their exposure to wildfire risk, and (to a lesser extent) changing neighborhood prices.

Disclosure induces systematic re-sorting based on risk preferences. As a result, the incidence of wildfire risk shifts across buyer types: mortgaged buyers of primary homes face less exposure, while cash buyers and buyers of secondary residences face correspondingly more.

We then use the estimated model to analyze alternative risk disclosure policy designs. We find that the

most effective policies successfully distinguish the highest-risk subset of neighborhoods, and that precision alone is not sufficient to improve the performance of a poorly-designed disclosure rule.

Together, these results carry a broader implication for climate risk policy. Information interventions do not eliminate exposure to environmental risk; they reallocate it across households through equilibrium adjustments in the housing market. In our context, that reallocation shifts exposure toward relatively advantaged households, which are less constrained by financing and insurance requirements, and therefore may be better positioned to bear it.

More broadly, our findings highlight the importance of incorporating information frictions and financial constraints into models of housing demand when evaluating climate risk policies. As governments increasingly rely on disclosure, insurance reform, and other information-based interventions to address climate risk, understanding how these policies interact with market structure will be essential for predicting their effects on both efficiency and equity.

References

Bakkensen, Laura A and Lint Barrage (2022) "Going underwater? Flood risk belief heterogeneity and coastal home price dynamics," *The Review of Financial Studies*, 35 (8), 3666–3709.

Bakkensen, Laura A. and Lala Ma (2020) "Sorting over Flood Risk and Implications for Policy Reform," *Journal of Environmental Economics and Management*, 104, 102362.

Barahona, Nano, Cristóbal Otero, and Sebastián Otero (2023) "Equilibrium effects of food labeling policies," *Econometrica*, 91 (3), 839–868.

Bartik, Timothy J (1991) *Who benefits from state and local economic development policies?*: WE Upjohn Institute for Employment Research.

Bass, Sarah and Michael Fogarty (2025) "Home Financing Responses to Wildfire Risk Information and Insurance Availability."

Baum-Snow, Nathaniel and Lu Han (2024) "The microgeography of housing supply," *Journal of Political Economy*, 132 (6), 1897–1946.

Bayer, Patrick, Fernando Ferreira, and Robert McMillan (2007) "A Unified Framework for Measuring Preferences for Schools and Neighborhoods," *Journal of Political Economy*, 115 (4), 588–638, 10.1086/522381.

Bayer, Patrick and Christopher Timmins (2007) "Estimating Equilibrium Models of Sorting Across Locations," *The Economic Journal*, 117 (518), 353–374, 10.1111/j.1468-0297.2007.02021.x.

Baylis, Patrick and Judson Boomhower (2023) "The Economic Incidence of Wildfire Suppression in the United States," *American Economic Journal: Applied Economics*, 15 (1), 442–473.

——— (2026) "Mandated versus Voluntary Adaptation to Natural Disasters: The Case of US Wildfires," *Journal of Political Economy*, 134 (3), 895–948, 10.1086/739331.

Berry, Steven, James Levinsohn, and Ariel Pakes (1995) "Automobile Prices in Market Equilibrium," *Econometrica*, 63 (4), 841–890, 10.2307/2171802.

Berry, Steven T (1994) "Estimating discrete-choice models of product differentiation," *The RAND Journal of Economics*, 242–262.

Bin, Okmyung and Craig E Landry (2013) "The Effects of Flood Risk on Housing Prices: Evidence from Recent Hurricane Events," *Journal of Environmental Economics and Management*, 65 (3), 361–376.

Biswas, Siddhartha, Mallick Hossain, and David Zink (2023) "California Wildfires, Property Damage, and Mortgage Repayment," Working Paper 23-05/R, Federal Reserve Bank of Philadelphia.

Boomhower, Judson and Meredith Fowlie (2026) "The Efficiency and Equity Impacts of Risk Classification in Catastrophe Insurance," *Environmental and Energy Policy and the Economy*, 7, 10.1086/738539.

Boomhower, Judson, Meredith Fowlie, Jacob Gellman, and Andrew Plantinga (2024) "How Are Insurance Markets Adapting to Climate Change? Risk Classification and Pricing in the Market for Homeowners Insurance," Working Paper 32625, National Bureau of Economic Research, 10.3386/w32625.

Boomhower, Judson, Meredith Fowlie, and Andrew J. Plantinga (2023) "Wildfire Insurance, Information, and Self-Protection," *AEA Papers and Proceedings*, 113, 310–315.

Brown, Claire (2025) "Zillow Removes Climate Risk Scores from Home Listings," *The New York Times*.

Collier, Benjamin, Tobias Huber, Johannes Gerd Jaspersen, and Andreas Richter (2023) "Homeowners' Willingness To Insure Flood Risks as Prices Increase," *Available at SSRN 4635177*.

Cook, Cody, Pearl Z Li, and Ariel J Binder (2023) *Where to build affordable housing?: Evaluating the tradeoffs of location*: US Census Bureau, Center for Economic Studies Rochester, NY.

Deryugina, Tatyana and Olga Shurchkov (2016) "The effect of information provision on public consensus about climate change," *PloS one*, 11 (4), e0151469.

Epple, Dennis, Thomas Romer, and Holger Sieg (2001) "Interjurisdictional sorting and majority rule: an empirical analysis," *Econometrica*, 69 (6), 1437–1465.

Epple, Dennis and Holger Sieg (1999) "Estimating equilibrium models of local jurisdictions," *Journal of political economy*, 107 (4), 645–681.

Fairweather, Daryl, Matthew E Kahn, Robert D Metcalfe, and Sebastian Sandoval-Olascoaga (2023) "The Impact of Climate Risk Disclosure on Housing Search and Buying Dynamics: Evidence from a Nationwide Field Experiment with Redfin."

Gibson, Matthew and Jamie T Mullins (2020) "Climate risk and beliefs in New York floodplains," *Journal of the Association of Environmental and Resource Economists*, 7 (6), 1069–1111.

Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift (2020) "Bartik instruments: What, when, why, and how," *American Economic Review*, 110 (8), 2586–2624.

Hansen, Bruce (2022) *Probability and statistics for economists*: Princeton University Press.

Hino, Miyuki and Marshall Burke (2021) "The effect of information about climate risk on property values," *Proceedings of the National Academy of Sciences*, 118 (17), e2003374118.

Ioannides, Yannis M and Jeffrey E Zabel (2008) "Interactions, neighborhood selection and housing demand," *Journal of urban economics*, 63 (1), 229–252.

Jaffe, Melissa R., Joe H. Scott, Michael N. Callahan, Gregory K. Dillon, Eva C. Karau, and Mitchell T. Lazarz (2024) "Wildfire Risk to Communities: Spatial Datasets of Wildfire Risk for Populated Areas in the United States," 10.2737/RDS-2020-0060-2.

Kousky, Carolyn (2010) "Learning from Extreme Events: Risk Perceptions after the Flood," *Land Economics*, 86 (3), 395–422.

Ma, Lala (2019) "Learning in a Hedonic Framework: Valuing Brownfield Remediation," *International Economic Review*, 60 (3), 1355–1387, https://doi.org/10.1111/iere.12389.

Ma, Lala, Margaret Walls, Matthew Wibbenmeyer, and Connor Lennon (2024) "Risk disclosure and home prices: Evidence from California wildfire hazard zones," *Land Economics*, 100 (1), 6–21.

Mulder, Philip (2024) "Mismeasuring risk: The welfare effects of flood risk information," *https://dx.doi.org/10.2139/ssrn.4966795*.

Mulder, Philip and Carolyn Kousky (2023) "Risk Rating without Risk Information," *AEA Papers and Proceedings*, 113, 316–320.

Sastry, Parinitha (2026) "Who Bears Flood Risk? Evidence from Mortgage Markets in Florida," *The Review of Financial Studies*, 39 (8), 2317–2357, 10.1093/rfs/hhag030.

Scott, Joe H., Gregory K. Dillon, Melissa R. Jaffe et al. (2024) "Wildfire Risk to Communities: Spatial Datasets of Landscape-Wide Wildfire Risk Components for the United States," 10.2737/RDS-2020-0016-2.

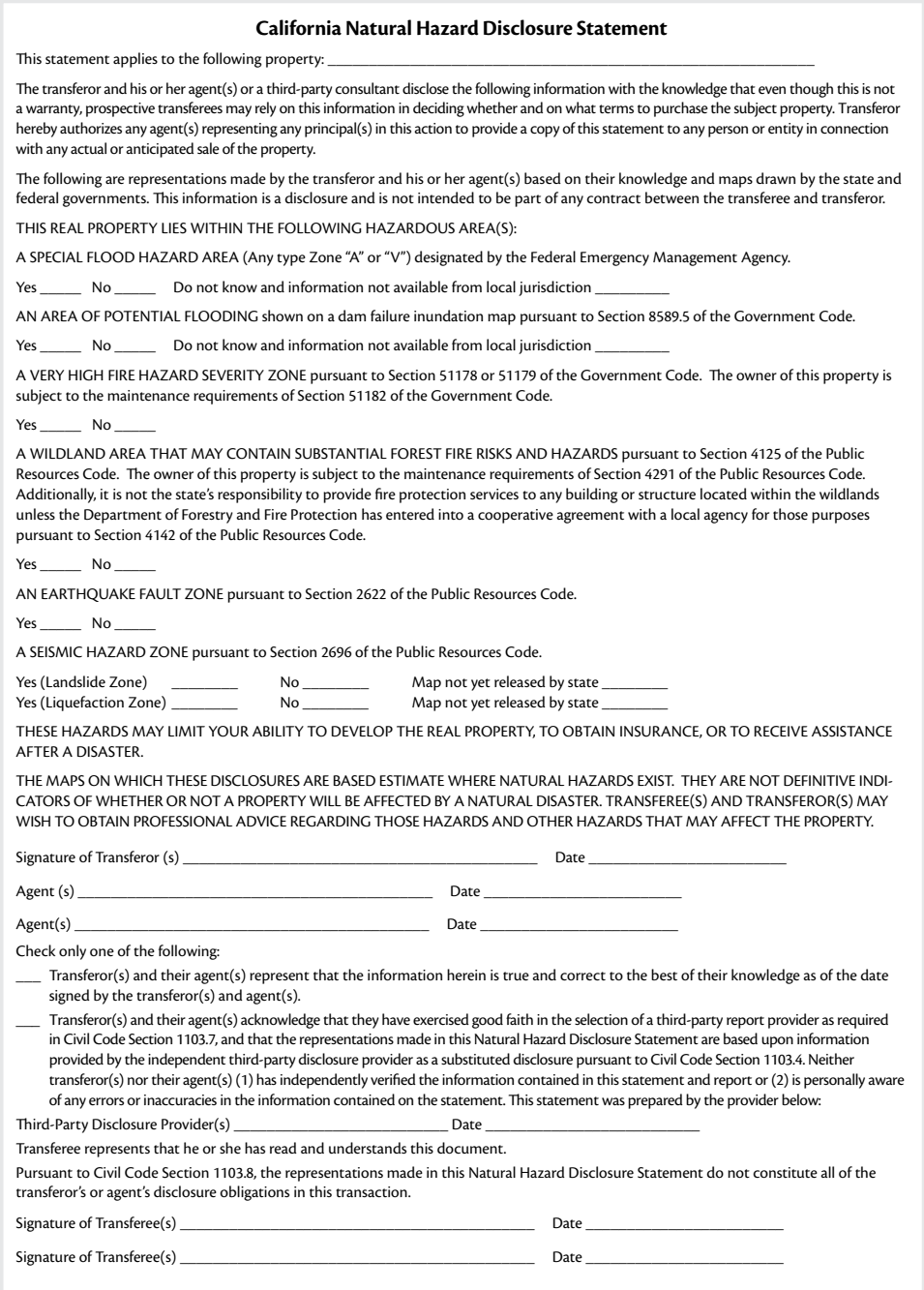
Sieg, Holger, V. Kerry Smith, H. Spencer Banzhaf, and Randy Walsh (2002) "Interjurisdictional Housing Prices in Locational Equilibrium," *Journal of Urban Economics*, 52 (1), 131–153, 10.1016/S0094-1190(02)00007-4.

Weill, Joakim A (2026) "Flood risk mapping and the distributional impacts of climate information," *American Economic Journal: Economic Policy*, 18 (3), 275–305.

Williams, A. Park, John T. Abatzoglou, Alexander Gershunov, Janin Guzman-Morales, Daniel A. Bishop, Jennifer K. Balch, and Dennis P. Lettenmaier (2019) "Observed Impacts of Anthropogenic Climate Change on Wildfire in California," *Earth's Future*, 7 (8), 892–910, 10.1029/2019EF001210.

Wright, Marvin N and Andreas Ziegler (2017) "ranger: A fast implementation of random forests for high dimensional data in C++ and R," *Journal of statistical software*, 77, 1–17.

A Additional Figures



California Natural Hazard Disclosure Statement

This statement applies to the following property: _____

The transferor and his or her agent(s) or a third-party consultant disclose the following information with the knowledge that even though this is not a warranty, prospective transferees may rely on this information in deciding whether and on what terms to purchase the subject property. Transferor hereby authorizes any agent(s) representing any principal(s) in this action to provide a copy of this statement to any person or entity in connection with any actual or anticipated sale of the property.

The following are representations made by the transferor and his or her agent(s) based on their knowledge and maps drawn by the state and federal governments. This information is a disclosure and is not intended to be part of any contract between the transferee and transferor.

THIS REAL PROPERTY LIES WITHIN THE FOLLOWING HAZARDOUS AREA(S):

A SPECIAL FLOOD HAZARD AREA (Any type Zone "A" or "V") designated by the Federal Emergency Management Agency.

Yes _____ No _____ Do not know and information not available from local jurisdiction _____

AN AREA OF POTENTIAL FLOODING shown on a dam failure inundation map pursuant to Section 8589.5 of the Government Code.

Yes _____ No _____ Do not know and information not available from local jurisdiction _____

A VERY HIGH FIRE HAZARD SEVERITY ZONE pursuant to Section 51178 or 51179 of the Government Code. The owner of this property is subject to the maintenance requirements of Section 51182 of the Government Code.

Yes _____ No _____

A WILDLAND AREA THAT MAY CONTAIN SUBSTANTIAL FOREST FIRE RISKS AND HAZARDS pursuant to Section 4125 of the Public Resources Code. The owner of this property is subject to the maintenance requirements of Section 4291 of the Public Resources Code. Additionally, it is not the state's responsibility to provide fire protection services to any building or structure located within the wildlands unless the Department of Forestry and Fire Protection has entered into a cooperative agreement with a local agency for those purposes pursuant to Section 4142 of the Public Resources Code.

Yes _____ No _____

AN EARTHQUAKE FAULT ZONE pursuant to Section 2622 of the Public Resources Code.

Yes _____ No _____

A SEISMIC HAZARD ZONE pursuant to Section 2696 of the Public Resources Code.

Yes (Landslide Zone) _____ No _____ Map not yet released by state _____
Yes (Liquefaction Zone) _____ No _____ Map not yet released by state _____

THESE HAZARDS MAY LIMIT YOUR ABILITY TO DEVELOP THE REAL PROPERTY, TO OBTAIN INSURANCE, OR TO RECEIVE ASSISTANCE AFTER A DISASTER.

THE MAPS ON WHICH THESE DISCLOSURES ARE BASED ESTIMATE WHERE NATURAL HAZARDS EXIST. THEY ARE NOT DEFINITIVE INDICATORS OF WHETHER OR NOT A PROPERTY WILL BE AFFECTED BY A NATURAL DISASTER. TRANSFEREE(S) AND TRANSFEROR(S) MAY WISH TO OBTAIN PROFESSIONAL ADVICE REGARDING THOSE HAZARDS AND OTHER HAZARDS THAT MAY AFFECT THE PROPERTY.

Signature of Transferor (s) _____ Date _____

Agent (s) _____ Date _____

Agent(s) _____ Date _____

Check only one of the following:

___ Transferor(s) and their agent(s) represent that the information herein is true and correct to the best of their knowledge as of the date signed by the transferor(s) and agent(s).

___ Transferor(s) and their agent(s) acknowledge that they have exercised good faith in the selection of a third-party report provider as required in Civil Code Section 1103.7, and that the representations made in this Natural Hazard Disclosure Statement are based upon information provided by the independent third-party disclosure provider as a substituted disclosure pursuant to Civil Code Section 1103.4. Neither transferor(s) nor their agent(s) (1) has independently verified the information contained in this statement and report or (2) is personally aware of any errors or inaccuracies in the information contained on the statement. This statement was prepared by the provider below:

Third-Party Disclosure Provider(s) _____ Date _____

Transferee represents that he or she has read and understands this document.

Pursuant to Civil Code Section 1103.8, the representations made in this Natural Hazard Disclosure Statement do not constitute all of the transferor's or agent's disclosure obligations in this transaction.

Signature of Transferee(s) _____ Date _____

Signature of Transferee(s) _____ Date _____

Figure A1: Natural Hazard Disclsosure Form

Notes:
This figure displays the updated Natural Hazard Disclosure (NHD) form mandated under California's 2021 wildfire risk disclosure policy. The fields indicating wildfire hazard and wildland areas with substantial forest fire risks are new as of 2021.

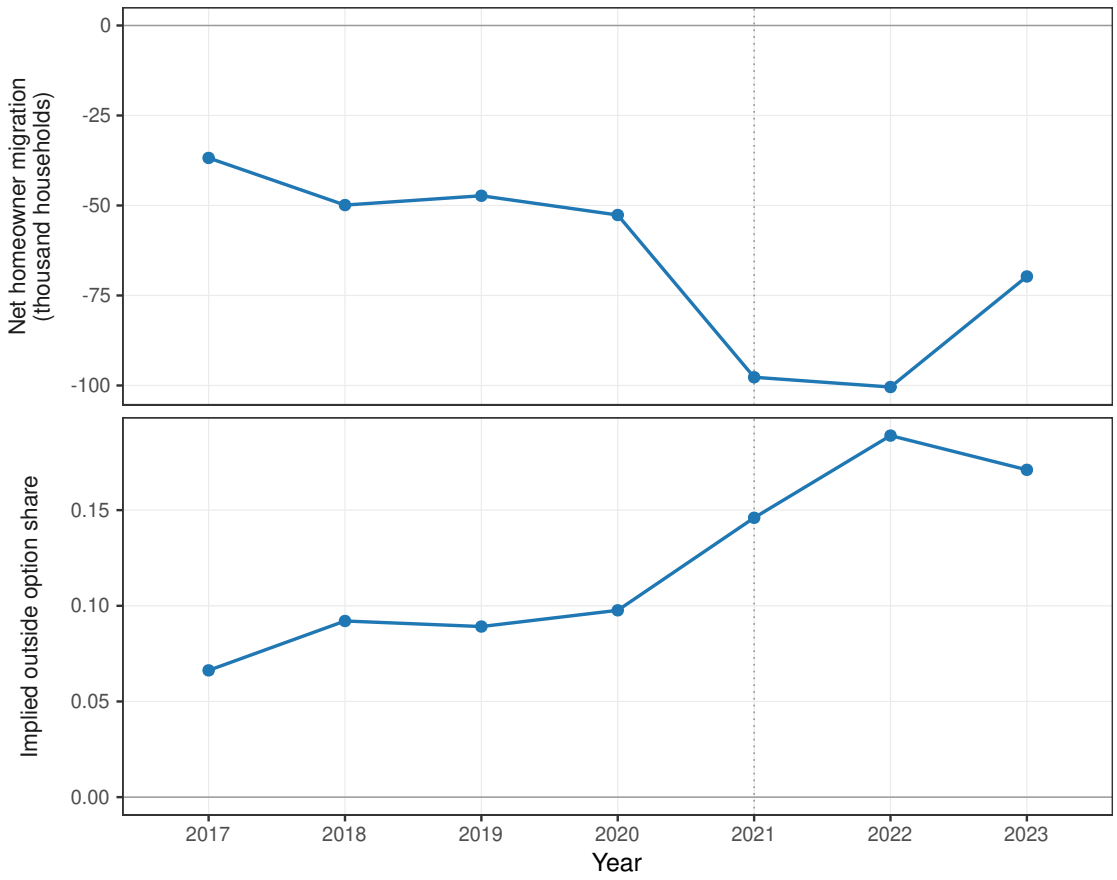


Figure A2: Estimated Out-migration and Implied Outside Option Share

Notes. The upper panel plots net homeowner migration for California: owner households whose householder moved into California from another state in the previous year, minus owner households whose householder moved from California to another state, with tenure measured at the destination. Counts are from the ACS 1-year PUMS, weighted by household weights. The ACS 1-year sample was not released for 2020, so 2020 scales net domestic migration from the Census Bureau Population Estimates Program by the average ratio of net owner households to net domestic migration in 2019 and 2021. The lower panel plots the implied outside-option share, $s_{0t} = M_t/N_t$, where M_t is the absolute value of net homeowner migration, $N_t = n_t + M_t$, and n_t is the number of in-state residential purchases in the CoreLogic data. s_{0t} is computed as one minus the sum of the neighborhood market shares, so it equals the outside share in the estimation sample. It rises when out-migration grows or when in-state sales fall. The dotted line marks 2021, the first year of the disclosure requirement.

B Additional Tables

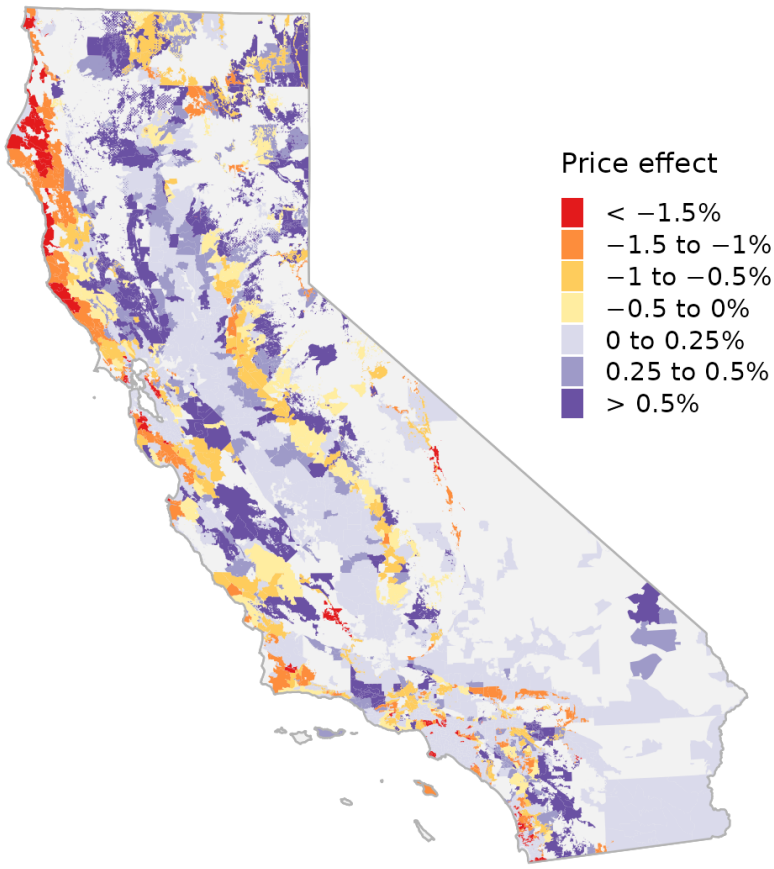


Figure A3: Estimated Out-migration and Implied Outside Option Share

Notes. Effect of the disclosure policy on neighborhood prices, $100 \times (\log P_{jt} - \log P_{jt}^{CF})$, the observed log price index minus its value in the counterfactual without disclosure, averaged over 2021–2023 within each neighborhood. Values are approximately percentages of home value; positive values mean disclosure raised prices. Estimated preferences and unobserved amenities are held fixed, with the pooled estimates of $\bar{\theta}^R$ and $\bar{\theta}^P$, and housing supply is perfectly inelastic. Areas outside the simulated neighborhoods are unfilled.

Table B1: Estimation Results: IV Decomposition

	OLS $\hat{\delta}_{jt}$	First stage $\log P_{jt}$	IV $\hat{\delta}_{jt}$
$\bar{\theta}^P$ log price $\log P_{jt}$	-0.572		-2.81
	(0.0351)		(0.269)
$\bar{\theta}^R$ belief $E[r_j \mid F_{jt}, D_j]$ (pp)	-1.22	0.338	-0.460
	(0.0324)	(0.0288)	(0.106)
W_{jt} shift-share instrument, standardized		-0.0271	
		(0.00239)	
Neighborhood fixed effects	Yes	Yes	Yes
First-stage F		92.9	
Within R^2	0.347	0.0504	0.262
Neighborhood-years	15,506	15,506	15,506

Notes. Regressions of the estimated mean utilities $\hat{\delta}_{jt}$ from the first step on log price and beliefs, with neighborhood fixed effects, pooling 2017–2023. Beliefs are the model-implied posterior mean burn probability $E[r_j \mid F_{jt}, D_j]$ at the first-step estimates, in percentage points. The first column is OLS; the second is the first stage, regressing log price on beliefs and the standardized shift-share instrument W_{jt}; the third is the IV estimate, instrumenting log price with W_{jt}. Standard errors, in parentheses, are clustered by neighborhood and treat the first-step estimates as known.

Table B2: No Disclosure Counterfactual: Alternative Supply Assumptions

	Fixed supply ($\eta_j = 0$)	Elastic supply (η_j, Baum-Snow–Han)	Perfectly elastic ($\eta_j \to \infty$)
Panel A: Prices (% of value)			
Disclosed ($D_j = 1$)	−0.65	−0.56	0
Not disclosed ($D_j = 0$)	0.14	0.13	0
Difference	−0.79	−0.69	0
Panel B: Sales (% change)			
Disclosed ($D_j = 1$)	0	−0.22	−1.78
Not disclosed ($D_j = 0$)	0	0.05	0.41
Panel C: Outside-option share (percentage points)			
All households	0	0.00	0.00
Mortgage, Primary	0.36	0.36	0.36
Mortgage, Non-Primary	−0.36	−0.35	−0.36
Cash, Primary	0.00	0.00	0.00
Cash, Non-Primary	−0.70	−0.69	−0.70
Panel D: Mean burn probability of buyers' neighborhoods (% change)			
All buyers	0	0.1	0.4
Mortgage, Primary	−9.4	−9.3	−9.3
Mortgage, Non-Primary	5.3	5.4	5.9
Cash, Primary	0.2	0.3	0.5
Cash, Non-Primary	8.5	8.7	9.4
Gap, Cash, Non-Primary vs. Mortgage, Primary	24.9	25.3	26.9
realized fire exposure	29.1	29.5	31.3
Panel E: Households who choose differently (%)			
All households	2.25	2.25	2.30
Mortgage, Primary	2.06	2.10	2.36
Mortgage, Non-Primary	2.21	2.18	1.95
Cash, Primary	0.50	0.45	0.18
Cash, Non-Primary	3.68	3.66	3.45

Notes. Effects of the disclosure policy under three housing-supply assumptions: the observed equilibrium minus the counterfactual without disclosure, in which beliefs evolve only with the fire history, pooled over 2021–2023 with years weighted by market size. Estimated preferences and unobserved amenities are held fixed, with $\bar{\theta}^R = -0.460$ and $\bar{\theta}^P = -2.81$ from the pooled IV. Fixed supply holds each neighborhood's sales at their observed level, elastic supply uses the elasticities described in Section 4.2.6 (sales-weighted mean 0.34; 0.55 in disclosed and 0.29 in not-disclosed neighborhoods), and perfectly elastic supply holds prices at their observed level. The outside-option share in the data is 16.7%. Belief changes do not depend on the supply assumption and are reported in Table 5. Prices are sales-weighted log-point differences ($\times 100$), and sales are the percent change in total sales in each group. Panel D reports the percent change in the buyer-weighted mean burn probability of the neighborhoods chosen; the gap rows are the percent change in the difference between Cash, Non-Primary and Mortgage, Primary buyers, measured by burn probability and by realized fire exposure from the year of purchase through 2023. Panel E is the minimum share of households, including those leaving California, who choose differently.

Table B3: No Disclosure Counterfactual, Pre-Period $\bar{\theta}^R$

	Fixed supply ($\eta_j = 0$)	Elastic supply (η_j, Baum-Snow–Han)	Perfectly elastic ($\eta_j \to \infty$)
Panel A: Prices (% of value)			
Disclosed ($D_j = 1$)	−6.03	−4.74	0
Not disclosed ($D_j = 0$)	0.36	0.72	0
Difference	−6.39	−5.46	0
Panel B: Sales (% change)			
Disclosed ($D_j = 1$)	0	−2.32	−13.68
Not disclosed ($D_j = 0$)	0	0.24	3.12
Panel C: Outside-option share (percentage points)			
All households	0	0.20	0.34
Mortgage, Primary	0.36	0.56	0.68
Mortgage, Non-Primary	−0.36	−0.13	0.06
Cash, Primary	0.00	0.00	0.00
Cash, Non-Primary	−0.70	−0.46	−0.24
Panel D: Mean burn probability of buyers' neighborhoods (% change)			
All buyers	0	−1.1	−6.2
Mortgage, Primary	−9.3	−10.4	−15.6
Mortgage, Non-Primary	5.4	4.2	−0.8
Cash, Primary	−1.0	−1.9	−6.3
Cash, Non-Primary	8.6	7.5	2.8
Gap, Cash, Non-Primary vs. Mortgage, Primary	25.0	24.1	20.4
realized fire exposure	29.3	28.2	24.7
Panel E: Households who choose differently (%)			
All households	2.23	2.27	2.80
Mortgage, Primary	2.03	2.37	4.33
Mortgage, Non-Primary	2.25	1.85	0.30
Cash, Primary	0.20	0.31	2.35
Cash, Non-Primary	3.73	3.32	1.14

Notes. Effects of the disclosure policy under three housing-supply assumptions: the observed equilibrium minus the counterfactual without disclosure, in which beliefs evolve only with the fire history, pooled over 2021–2023 with years weighted by market size. Estimated preferences and unobserved amenities are held fixed, with $\bar{\theta}^R = -0.791$ from the IV on 2017–2020 and $\bar{\theta}^P = -2.81$ from the pooled IV. Fixed supply holds each neighborhood's sales at their observed level, elastic supply uses the elasticities described in Section 4.2.6 (sales-weighted mean 0.34; 0.55 in disclosed and 0.29 in not-disclosed neighborhoods), and perfectly elastic supply holds prices at their observed level. The outside-option share in the data is 16.7%. Belief changes do not depend on the supply assumption and are reported in Table 5. Prices are sales-weighted log-point differences ($\times 100$), and sales are the percent change in total sales in each group. Panel D reports the percent change in the buyer-weighted mean burn probability of the neighborhoods chosen; the gap rows are the percent change in the difference between Cash, Non-Primary and Mortgage, Primary buyers, measured by burn probability and by realized fire exposure from the year of purchase through 2023. Panel E is the minimum share of households, including those leaving California, who choose differently.

Table B4: Alternative Disclosure Designs: Full Results

	Binary label				Decile score	
	Observed		From map		From map	
	Noisy	Precise	Noisy	Precise	Noisy	Precise
Panel A: Information						
Prior variance resolved (%)	12.2	16.3	12.2	16.3	17.8	64.1
Mean absolute belief change (pp)	0.096	0.109	0.111	0.126	0.099	0.055
Panel B: Beliefs, by observed designation (change, pp)						
Disclosed ($D_j = 1$)	0.450	0.504	0.343	0.372	0.197	0.104
Not disclosed ($D_j = 0$)	−0.017	−0.021	0.038	0.047	0.069	0.003
Panel C: Prices, by observed designation (% of value)						
Disclosed ($D_j = 1$)	−0.65	−0.80	−0.42	−0.55	−0.19	1.05
Not disclosed ($D_j = 0$)	0.14	0.16	0.00	0.00	−0.22	0.02
Difference	−0.79	−0.96	−0.41	−0.55	0.04	1.04
Panel D: Outside-option share (percentage points)						
Mortgage, Primary	0.36	0.40	0.50	0.56	0.47	0.07
Mortgage, Non-Primary	−0.36	−0.40	−0.51	−0.58	−0.51	−0.01
Cash, Primary	0.00	0.00	0.00	0.00	0.00	0.00
Cash, Non-Primary	−0.70	−0.77	−0.95	−1.08	−0.87	−0.20
Panel E: Mean burn probability of buyers' neighborhoods (% change)						
Mortgage, Primary	−9.4	−8.5	−11.0	−10.5	−8.7	−11.6
Mortgage, Non-Primary	5.3	5.0	6.3	6.3	3.9	3.3
Cash, Primary	0.2	0.7	0.8	1.4	−1.0	−4.9
Cash, Non-Primary	8.5	7.2	9.4	8.3	8.9	16.2
Gap between extreme types	24.9	21.5	28.3	25.6	25.1	41.8
Panel F: Households who choose differently (%)						
All households	2.25	2.54	2.68	3.07	1.50	0.85
Mortgage, Primary	2.06	2.33	2.46	2.82	1.37	0.74
Mortgage, Non-Primary	2.21	2.51	2.67	3.07	1.48	0.61
Cash, Primary	0.50	0.56	0.62	0.70	0.41	0.35
Cash, Non-Primary	3.68	4.14	4.34	4.96	2.41	1.71

Notes. Effects of alternative disclosure designs: each design minus the counterfactual without disclosure, in which beliefs evolve only with the fire history, pooled over 2021–2023 with years weighted by market size. The binary label either keeps the observed designations (observed) or assigns them from the map (from map), designating neighborhoods with $R_j > e^{\bar{r}}$; the decile score assigns each neighborhood its decile of R_j among California neighborhoods. Households read a label either with the noise of the estimated disclosure rule (noisy) or as revealing the range that contains r_j (precise). The noisy binary label with observed designations is the actual policy. Supply is fixed, and estimated preferences and unobserved amenities are held fixed, with $\bar{\theta}^R = -0.460$ and $\bar{\theta}^P = -2.81$ from the pooled IV. Prior variance resolved is the share of the prior variance of r_j that the design resolves before households observe their labels, $\mathrm{Var}(E[r_j \mid S_j])/\mathrm{Var}(r_j)$, a property of the signal structure. Beliefs and prices group neighborhoods by their observed designation in every column. Prices are sales-weighted log-point differences ($\times 100$). Panel E reports the percent change in the buyer-weighted mean burn probability of the neighborhoods each type chooses; the gap between extreme types is the percent change in the difference between Cash, Non-Primary and Mortgage, Primary buyers, the least and most risk-sensitive types. Panel F is the minimum share of households, including those leaving California, who choose differently.

C Belief Updating on Disclosure

This appendix derives the posterior mean in equation (8), gives its closed form under the Gamma prior, and describes how the updating factor varies across neighborhoods. Throughout, E_t, $\Pr_t$, and Cov_t denote moments under the fire-updated prior $r_j \sim \Gamma(a_{jt}, \beta_t)$, with shape $a_{jt} = \kappa R_j + F_{jt}$ and rate $\beta_t = \kappa + T_t$, and $\mu_{jt} = a_{jt}/\beta_t$ is the prior mean.

Posterior mean. Let $L_j(r) = \Pr(D_j \mid r_j = r)$ denote the likelihood of the observed designation. By Bayes' rule,

$$E[r_j \mid F_{jt}, D_j] = \frac{E_t[r_j\, L_j(r_j)]}{E_t[L_j(r_j)]}.$$

The denominator is the prior probability of the designation, $\Pr_t(D_j)$. Writing the numerator as $E_t[r_j]\, E_t[L_j(r_j)] + \mathrm{Cov}_t(r_j, L_j(r_j))$ gives

$$E[r_j \mid F_{jt}, D_j] = \mu_{jt}\left(1 + \frac{\mathrm{Cov}_t\left(r_j,\, L_j(r_j)\right)}{\mu_{jt}\, \Pr_t(D_j)}\right),$$

which is equation (8). This decomposition holds for any prior and any likelihood. Because $\Pr(D_j = 1 \mid r_j)$ is increasing in r_j, the covariance is positive after a disclosure and negative after non-disclosure. The two posterior means average back to the prior, $\Pr_t(D_j = 1)\, E[r_j \mid F_{jt}, D_j = 1] + \Pr_t(D_j = 0)\, E[r_j \mid F_{jt}, D_j = 0] = \mu_{jt}$, so the designation moves beliefs in both directions but not on average.

Closed form under the Gamma prior. Let

$$q(a) \equiv \int_0^\infty \Phi\left(\frac{\log r - \bar{r}}{\tau}\right) g(r;\, a,\, \beta_t)\, \mathrm{d}r$$

denote the probability of disclosure when $r_j \sim \Gamma(a, \beta_t)$, so that $\Pr_t(D_j = 1) = q(a_{jt})$. The numerator of the posterior mean weights each value of r_j by r_j itself. Weighting a Gamma density by r yields another Gamma density with the shape increased by one,

$$r\, g(r;\, a, \beta) = \frac{a}{\beta}\, g(r;\, a + 1, \beta),$$

so $E_t[r_j \Pr(D_j = 1 \mid r_j)] = \mu_{jt}\, q(a_{jt} + 1)$. The posterior means are therefore

$$E[r_j \mid F_{jt}, D_j = 1] = \mu_{jt} \frac{q(a_{jt} + 1)}{q(a_{jt})},$$
$$E[r_j \mid F_{jt}, D_j = 0] = \mu_{jt} \frac{1 - q(a_{jt} + 1)}{1 - q(a_{jt})}.$$

The distribution $\Gamma(a + 1, \beta)$ first-order stochastically dominates $\Gamma(a, \beta)$, and $\Pr(D_j = 1 \mid r_j)$ is increasing in r_j, so $q(a_{jt} + 1) > q(a_{jt})$. This corresponds to the sign of the covariance term in the more general formulation above.

Comparison with normal beliefs. As in Barahona et al. (2023), the effect of disclosure is nonlinear in the distance between the prior and the disclosure threshold. Measured in logs, the update after a disclosure grows with the distance between $\log \mu_{jt}$ and $\bar{r}$, as in their model. Measured in levels, however, it is bounded, because disclosed neighborhoods with low prior risk converge to nearly the same belief.

Computation. The disclosure probability is effectively flat outside $\log r \in [\bar{r} - 8\tau,\ \bar{r} + 8\tau]$. We evaluate $q(a_{jt})$ and $q(a_{jt} + 1)$ exactly on the two tails using incomplete Gamma functions, and by numerical integration on a fixed grid in $\log r$ over the interior.

D Hedonic Price Indices

To recover the time-varying neighborhood prices P_{jt}, we estimate the following hedonic regression:

$$\ln p_{kt} = a_{j(k),t} + H_k\Gamma + u_{kt}$$

We estimate the hedonic model on CoreLogic housing transaction data for the state of California from 2017 through Q2 of 2025. We restrict our sample to arms-length transactions of single-family residences, which leaves us with about 3.8 million transactions over the nine-year period. To enrich the hedonic model, we link the transaction data to the CoreLogic property file to obtain rich data on the characteristics of the properties that are sold and the structures on the property.

The goal is to recover estimates of a_{jt}, the neighborhood-time fixed effects. Following Sieg et al. (2002), these fixed effects can be used as the log prices in the sorting model.

Table D1 shows the estimates the hedonic model. In general, coefficients on the continuous covariates all take the expected sign, except for stories, which is negatively associated with price (conditional on square footage, number of bedrooms, and number of bathrooms).

Our goal in estimating the hedonic model is to obtain the best estimates of the ZIP-year fixed effects. This is distinct from the usual case, where the high-dimensional fixed effects are treated as nuisance parameters and the objects of interest are the coefficients on the "normal" variables in the model. To this end, we make a number of specification choices that are designed to maximize the number of observations that we can use to compute the fixed effects, particularly with respect to our treatment of variables with missing values for some of the characteristics H_k.

For categorical variables (such as building quality or construction type), we simply encode missingness as an additional level of the variable, and the model estimates a distinct intercept for these observations. For continuous covariates (such as building square footage and lot acreage), we first replace missing observation with a common stand-in value, which in practice we implement as the mean of the non-missing values. We also create an auxiliary indicator variable for cases with missing values of that covariate, and include both the original variable with the missing values filled in and the missing value dummy in the model.[18]

[18] Incidentally, both of these procedures for including observations with missing values in the RHS variables tend to *reduce* the R^2 of the hedonic model. This is due to the fact that including extra observations increases the total sum of squares in the dependent variable (log price) without adding much in the way of increasing the explanatory power of the model. If our

Table D1: Hedonic Regression Results

Dependent Variable:	ln(Sale Amount)
Model:	(1)
Variables	
ln(Square Footage)	0.5421***
	(0.0071)
ln(Acres)	0.0268***
	(0.0028)
ln(Stories)	-0.1140***
	(0.0053)
ln(Bedrooms)	0.0754***
	(0.0062)
ln(Bathrooms)	0.0811***
	(0.0052)
1(Manufactured Home)	-0.3991***
	(0.0225)
1(Pool)	0.0735***
	(0.0050)
Fixed-effects	
ZIP x Year	Yes
Quarter	Yes
Decade Built	Yes
Building Quality	Yes
Garage Type	Yes
Construction Type	Yes
Fit statistics	
Observations	3,811,646
R^2	0.64829
Within R^2	0.17986

Clustered (ZIP x Year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Figure D1 plots the estimated ZIP-year fixed effects against the simple within-cell means of the log transaction prices using a binned scatterplot. The blue line overlaid on the data is the best-fit linear regression estimate, while the black line is the identity. The slope of best-fit line is about 0.85, which demonstrates that including the structural characteristics of the housing stock in the hedonic model attenuates the relationship between the mean price and the estimated fixed effects. This suggests that areas with higher housing costs (as measured by the fixed effects a_{jt}) tend to have higher-valued structural features, such as larger lots, more square footage, and more bedrooms and bathrooms. This illustrates the importance of using the hedonic model to estimate the a_{jt} parameters; otherwise in the sorting model we would erroneously attribute to the neighborhood amenities ξ_j variation in prices P_jt that is explained by the characteristics of housing in a a given neighborhood.

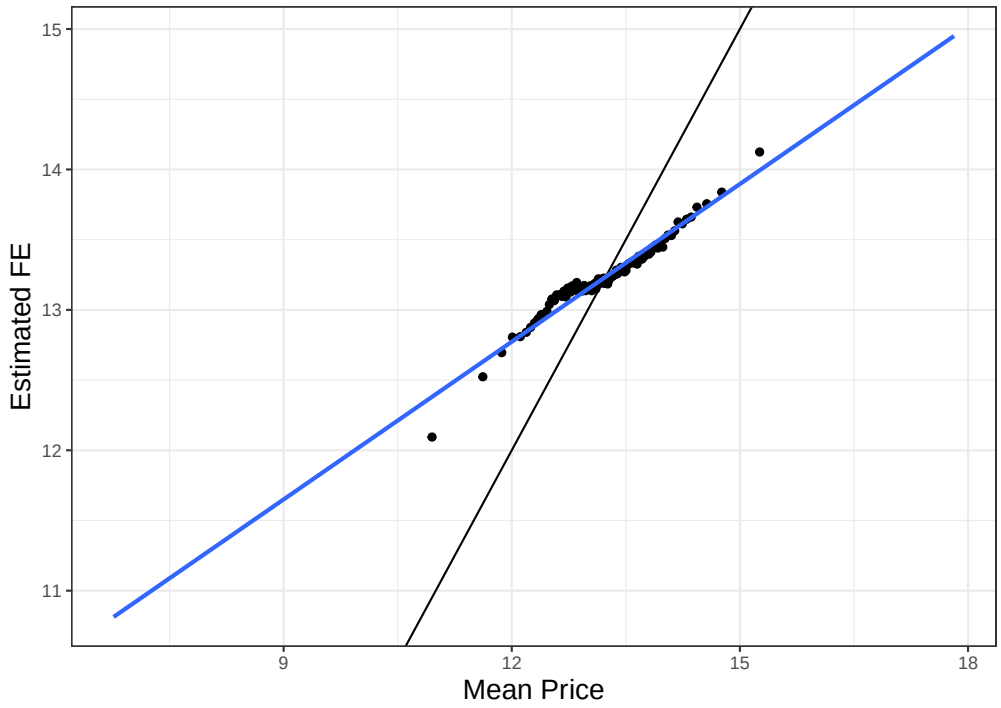


Figure D1: Fixed Effects Estimates

Figure D2 compares the (log) distribution of standard errors from the hedonic model (in red) and the standard errors of the ZIP-year cell means (in blue). The standard errors from the hedonic model are about 25% smaller than the standard errors of the means, which illustrates the advantage of including the rich set of housing characteristics for estimating the fixed effects a_{jt}.

goal was to maximize the model R^2 or obtain consistent estimates of the Γ parameters, this would be a problem. However, for our particular aim of obtaining the most precise estimates possible of the a_{jt} parameters, the fact that adding these additional

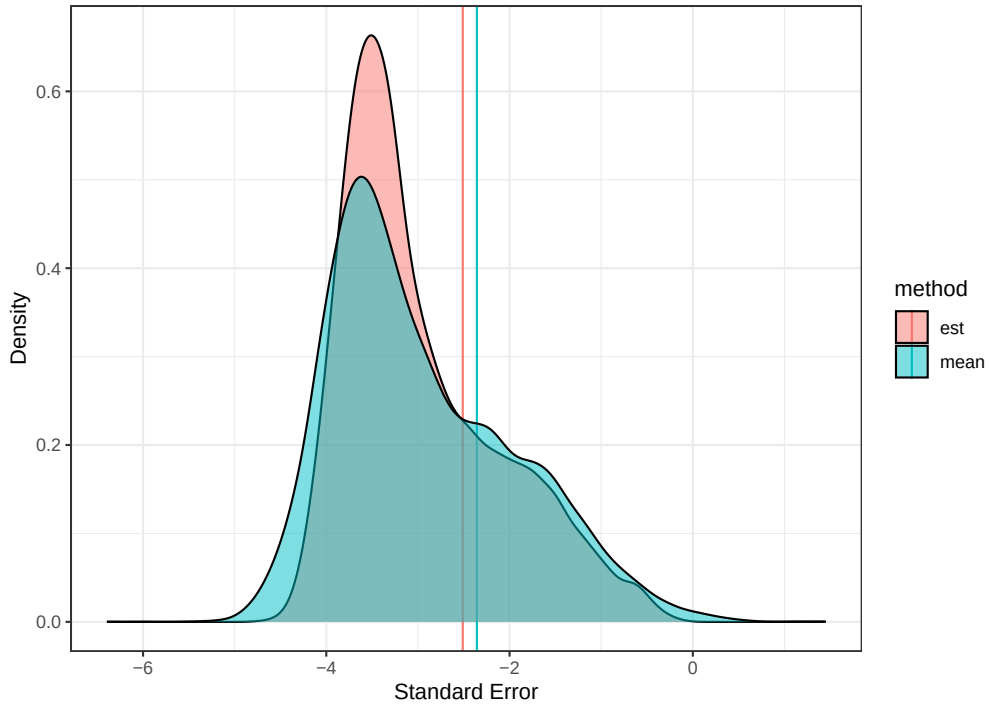


Figure D2: Fixed Effect Standard Error Comparison

D.1 Random Forest

To more flexibly estimate the hedonic model, we generalize the linear model above into the following form:

$$\ln p_{kt} = a_{j(k),t} + f(H_k) + e_{kt}$$

To estimate the model, we proceed in two steps, applying the logic of the FWL theorem to first flexibly estimate the effects of the structural characteristics H_k on prices p_{kt} using a random forest model (Wright and Ziegler, 2017) to estimate the hedonic function $f(\cdot)$.

$$\ln p_{kt} = f(H_k) + u_{kt}$$

Then, having estimated the function $\hat{f}(\cdot)$ via random forest, we construct the first-step residuals:

$$\hat{u}_{kt} = \ln p_{kt} - \hat{f}(H_k)$$

observations increases the number of observations in each ZIP-year cell improves precision.

To recover the zip-year fixed effects, we then run the following regression:

$$\hat{u}_{kt} = a_{j(k),t} + e_{kt}$$

Using default settings, the random forest already substantially outperforms the linear model, and after tuning the model hyperparameters, model performance (as assessed by RMSE or R^2) increases further.

D.2 Fixed Effects Decomposition

We also perform a decomposition exercise where we regress the estimated ZIP-year fixed effects on separate zip code and year fixed effects to determine the variance explained by each component (and the residual variance left that is not independently explained by either the time or location FE).

$$a_{jt} = a_j + \tau_t + e_{jt}$$

The model R^2 is about 0.87, which suggests that place and time fixed effects explain most of the variation in the fixed effects from the hedonic model. Of that 0.87, about 0.27 can be attributed to the year fixed effects, while the remaining 0.60 can be attributed to the ZIP fixed effects.

Figure D3 plots the point estimates of the year fixed effects, which can roughly be interpreted as percent growth in average house prices in California compared to a base year of 2017 (the omitted fixed effect category in the model). The estimates suggest that prices grew somewhat modestly across the state from 2017-2020, before growing rapidly between 2020-2022 and leveling off since 2022. These estimates are very consistent with other house price indices, such as the FHFA All-Transactions House Price Index for California.[19]

Figure D4 maps the estimated ZIP fixed effects for the state of California. As expected, prices are higher along the coast and in the major metro areas, and lower inland and in more rural areas. Areas in the map without estimated fixed effects correspond to federal landholdings, which tend to be vacant or extremely sparsely populated.

[19] FRED series `CASTHPI`

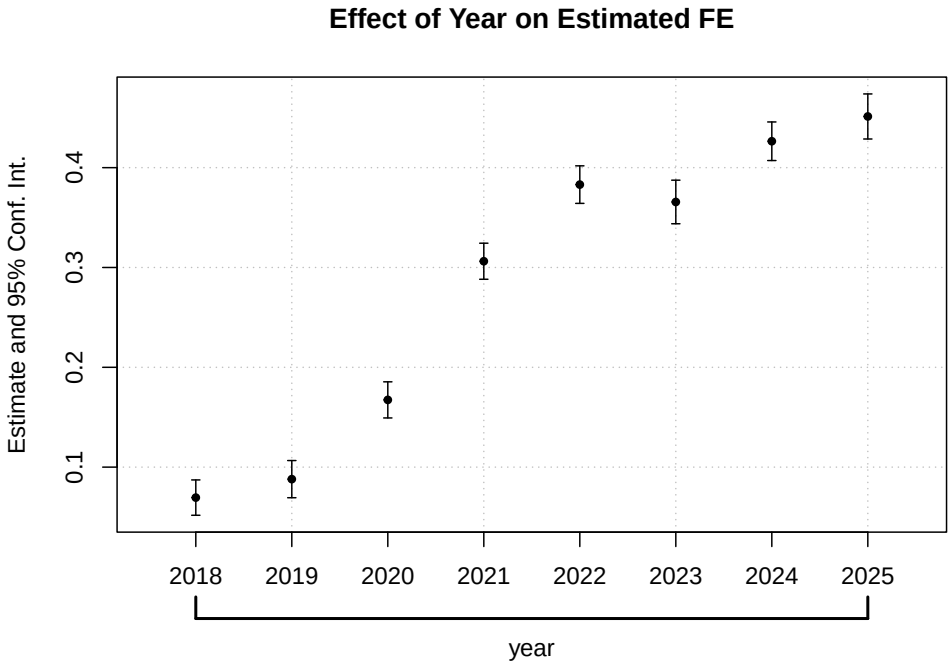


Figure D3: Year Fixed Effects

E Estimation Details

E.1 Numerical Optimization

We solve the maximization problem using the Newton algorithm. Starting from an initial guess $\boldsymbol{\theta}^0 = \mathbf{0}$, we iterate the update rule:

$$\boldsymbol{\theta}^{h+1} = \boldsymbol{\theta}^h - \alpha H(\boldsymbol{\theta}^h)^{-1} \nabla \ell(\boldsymbol{\theta}^h)$$

The step size $\alpha \in (0, 1]$ is chosen by a backtracking line search to prevent overshooting. We iterate until $\|\nabla \ell(\boldsymbol{\theta})\| < \epsilon$, with $\epsilon = \texttt{1e}{-}\texttt{8}$. Due to the global concavity of the log-likelihood function, the Newton algorithm converges to the unique global optimum. Results are robust to using derivative-free optimization methods as a check.

E.2 Standard Errors

We follow Hansen (2022, sections 10.15, 10.19) and consider three variance-covariance estimators:

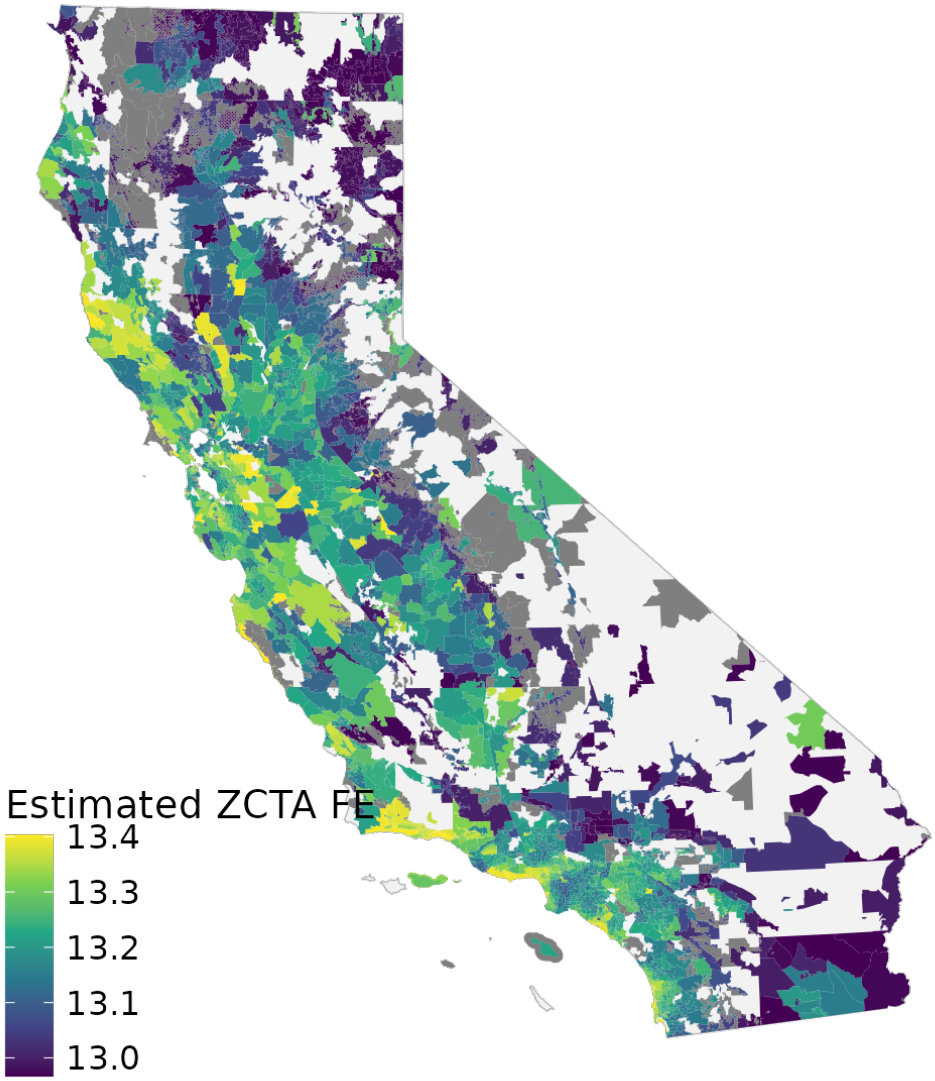


Figure D4: ZIP Fixed Effect Map

i. $\widehat{V}_1 = \left(-\frac{1}{N}H(\hat{\boldsymbol{\theta}})\right)^{-1}$, the plug-in Hessian estimator.

ii. $\widehat{V}_2 = \left(\frac{1}{N}\mathcal{I}(\hat{\boldsymbol{\theta}})\right)^{-1}$, where $\mathcal{I}(\boldsymbol{\theta})$ is the Fisher information matrix.

iii. $\widehat{V}_3 = \left(-\frac{1}{N}H(\hat{\boldsymbol{\theta}})\right)^{-1}\left(\frac{1}{N}\mathcal{I}(\hat{\boldsymbol{\theta}})\right)\left(-\frac{1}{N}H(\hat{\boldsymbol{\theta}})\right)^{-1}$, the robust sandwich estimator.

Our preferred estimator is $\widehat{V}_3$, which is robust to model misspecification. Standard errors are obtained as $\text{se}(\hat{\theta}) = \sqrt{\text{diag}(\widehat{V}_3)/N}$.